Introduction

Nothing is more crucial today than energy, Since the lack of energy represents a major barrier to the current society, including insufficient food, warm housing, and Internet access, as well as the use of nanotechnology products, the emergence of contemporary tendencies, potential new technology for meeting energy demands even brand-new physical and chemical techniques for the creation and management of effective systems to produce, gather, transform, and energy transfer into its many forms, Eventually, if current trends continue, civilization will demand more electrical power. I believe a strong likelihood that greenhouse gas emissions will significantly increase global warming over the 50 years from now. Climate change is a legitimate concern. It is the most pressing threat that is now prevailing. affecting the entire species, thus we must cooperate. collectively, put a stop to it. For our family and business, solar energy is a greener, safer investment. We can quickly lower your energy expenditures, give you energy independence from rising energy prices and rising carbon dioxide (CO_2) emissions, and raise the market value of your home or structure.

Utilizing solar energy has been an aim in the scientific community ever since the Becquerel identified the first photovoltaic effect in 1839. The amount of solar energy that is absorbed by Earth's atmosphere each hour is more than enough to meet all of the world's energy needs for a full year. Due to this, research into the most effective solar cells has expanded in recent years in an effort to wean the globe off of its reliance on fossil fuels. The advantages of PV technology are numerous and substantial. Everywhere in the planet, there is access to solar energy, which is renewable.

Unlike many other power generation methods, solar PV systems are portable, compact, and extremely modular. Solar photovoltaic (PV) energy has no fuel expenses and relatively low operation and maintenance (O&M) costs, in contrast to traditional power plants using coal, nuclear, oil, and gas. PV can therefore create energy indefinitely (as long as the sun is shining). A completely sustainable and eco-friendly way to generate energy is through photovoltaics. Additionally, it is free, so why not make use of it?[1].

Photovoltaics (PV) is a term used by scientists to describe solar cells, which are the smallest component of solar technology. Photovoltaics is the process of directly converting sunlight into electricity using solar cells. PV devices (solar cells) are unusual in that they convert incident solar energy directly into electricity with no noise, pollution, or moving components, making them durable, reliable, and long-lasting. Solar cells are built on the same principles and materials that drove the communications and computer revolutions. Today, it is a quickly developing and increasingly essential renewable alternative to conventional fossil fuel energy generation, although it is a relative newbie when compared to other electricity generating technologies, with the first viable photovoltaic systems exhibited in the 1950s.

In this report, the efficiency as well as other factors affecting it’s performance will be explained through the method of impedance spectroscopy, Impedance (Z), which measures how well a circuit can withstand current, is quite similar to resistance in this regard. Resistance is a concept for perfect resistors, but because many circuits are more complicated (they don't perfectly obey Ohm's Law, are frequency independent, and don't exhibit a phase shift between the current and voltage signals), impedance is used in place of resistance. Impedance accounts for all the factors that are confined to an ideal resistor in addition to other elements like inductance, resistance, and capacitance. Electrochemical impedance spectroscopy (EIS) measures the electrical current flowing through a sample when an AC voltage is applied at various frequencies. The frequency-dependent potential to frequency-dependent current ratio, or impedance (Z), can then be computed.

It is possible to measure frequencies numerous times using this method. It can be used to examine the concurrent operation of multiple electrochemical processes, the rate at which electrons are transferred during a reaction, diffusion-limited reactions, or the capacitive behavior of a system. EIS is used for a variety of tasks, such as identifying food ageing, measuring bacterial concentration in label-free biosensors, and researching ion mobility in batteries and super capacitors.

The identification of specific electrical components of molecular junctions and faultless contact resistance values are made possible by electrical impedance spectroscopy (EIS), which works in conjunction with a circuit model to make it easier to represent actual transport mechanisms.

Solar spectrum

The electromagnetic wave that makes up light has a frequency of about 1015 Hz. Numerous light frequencies are referred to as a "spectrum" in scientific terminology. In addition to a rainbow of colours and some light that cannot be seen with the human eye, the sun's white (or yellowish) light also contains other colours (ranging from blue to ultraviolet and in other direction from red to infrared). Therefore, we aim for the maximum energy absorption while designing a solar cell. But the maximum amount of energy from a basic cell (single pn junction) that may be converted into electricity is only 33%[1] due to physical restrictions. And that's not even mentioning how challenging it is to reach this amount. The efficiency of a single junction solar cell is only about 25.8% because of optical losses, imperfections in the materials, and manufacturing flaws.

Generations of solar cells

The solar cells are usually categorized into 3 generations, they are:

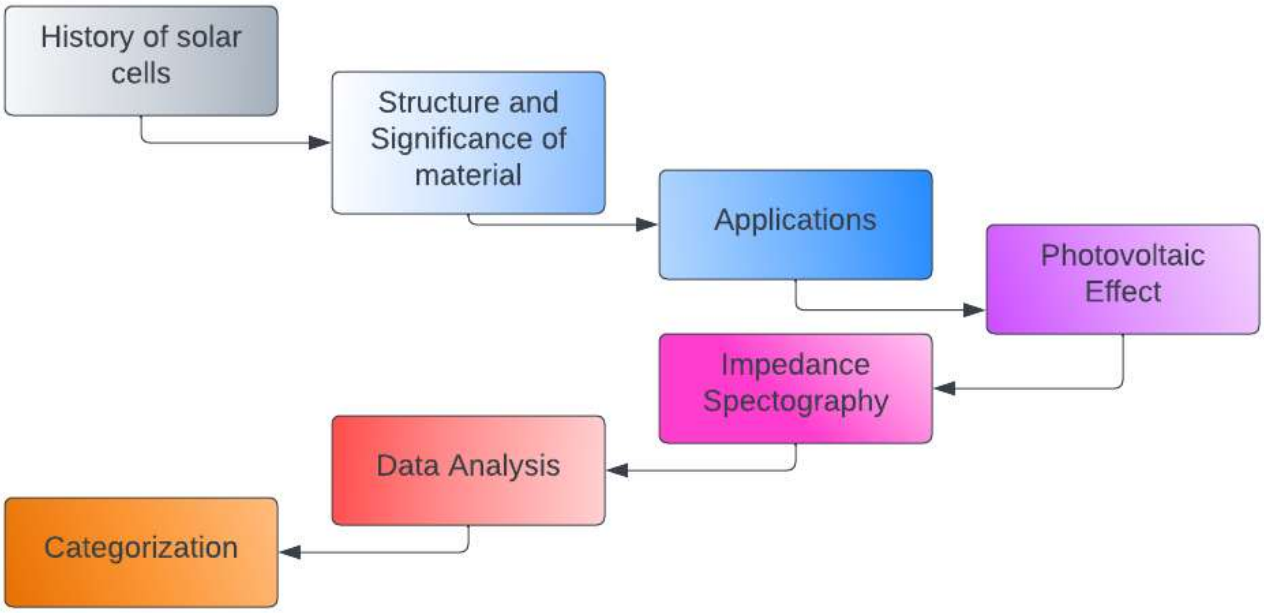
- First-generation solar cells have an average performance of 15-20% efficiency and are mostly based on silicon technology.
- Amorphous silicon, CIGS, or CdTe-based second-generation solar cells currently have low efficiency.
- Organic components are present in third-generation solar cells, which have a 30% efficiency potential. Recently, perovskite solar cell researchers attained efficiencies of more than 20%. [2].

Aim and objectives

The aim of this experiment is to analyze different solar cells, and with the help of Impedance spectroscopy data analysis, provide which cells are more efficient than others. The objectives of the experiment are,

- Study the characteristics of different solar cells,
- With the help of the tool, Impedance spectroscopy, analyze different parameters of the solar cell,
- Analyze the electric behavior of the cell under external applied voltage (in dark) and under illumination,
- Establish a figure of merit for these cells and categorize them under good, bad or ugly.

Literature Review :



History of solar cells:

The first solar cells were made from selenium in 1883 by an American inventor named Charles Fritts. Fritts had hoped that his solar cells would be able to rival Edison's coal-fired power plants, but they were only 1% effective in converting sunlight into electricity, making them unworkable. Despite the fact that some uses for selenium photovoltaics were discovered and some study on them persisted for several decades, they were not generally utilised. [1].

Richard Shoemaker In 1940, Ohl, a Bell Labs semiconductor researcher, developed the next major advancement in solar cell technology. One of the silicon samples he had been examining had a break in the middle. He discovered that this particular sample conducted current when it was exposed to light. The boundary between two sections with different levels of impurities was marked by a fracture, which most likely formed when the sample was made. As a result, one side was positively and the other negatively doped. Ohl had unintentionally produced a p-n junction, the key component of a solar cell. An electric field is created when excess positive charge builds up on one side of the p-n barrier and excess negative charge builds up on the other. When the cell is wired into a circuit, an incoming photon has the power to jolt an electron and start current flowing. Ohl patented his solar cell, which had an efficiency of only about 1%. [2]. The first functional silicon solar cell was created by a group of scientists at Bell Labs thirteen years later. In 1953, Daryl Chapin, an engineer who had previously worked at Bell Labs on magnetic materials, was tasked with finding a way to power telephone lines in far-off, moist locations where dry cell batteries degraded too quickly. The most promising alternative energy source, according to Chapin's research, is solar power. Selenium solar cells were the subject of his experiments, but they were ineffective.

The first functional silicon solar cell was created by a group of scientists at Bell Labs thirteen years later. In 1953, Daryl Chapin, an engineer who had previously worked at Bell Labs on magnetic materials, was tasked with finding a way to power telephone lines in far-off, moist locations where dry cell batteries degraded too quickly. The most promising alternative energy source, according to Chapin's research, is solar power. Selenium solar cells were the subject of his experiments, but they were ineffective. [3].

The first functional silicon solar cell was created by a group of scientists at Bell Labs thirteen years later. In 1953, Daryl Chapin, an engineer who had previously worked at Bell Labs on magnetic materials, was tasked with finding a way to power telephone lines in far-off, moist locations where dry cell batteries degraded too quickly. The most promising alternative energy source, according to Chapin's research, is solar power. Selenium solar cells were the subject of his experiments, but they were ineffective. Additionally, researchers found that the boron-arsenic silicon sold allowed them to make trustworthy electrical contacts. After a few more design changes, they joined many solar cells to create what they called a "solar battery." The initial silicon solar cells were around 6% more efficient at converting solar energy into electricity than older solar cells. [4].

Rooftop solar panels and pocket calculators are only two examples of the modern technology that uses solar cells. In order to make solar power more competitive with fossil fuels, research and development are being done with the goal of reducing the cost and raising the efficiency. Solar cells with an efficiency of more than 40% may now be produced thanks to advanced designs and materials. [5].

Structure and Significance of material :

Structure:

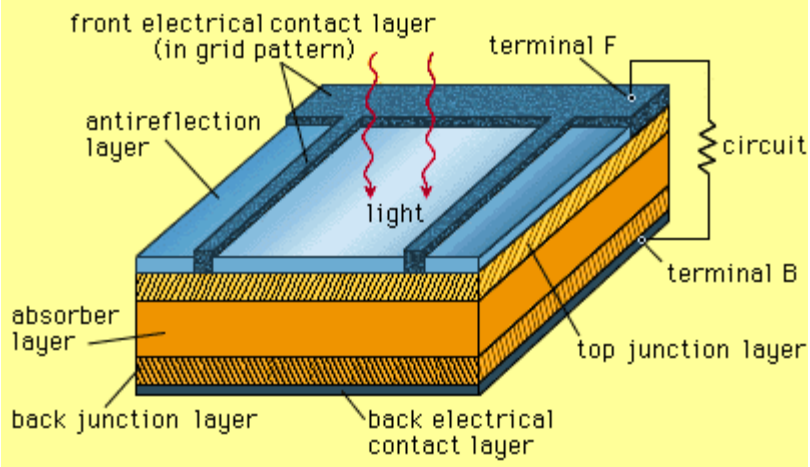


Figure 1 Structure of a solar cell

Solar cells can be found in spacecraft, central power plants, and calculators, but their basic design is the same in each case. Light can enter the device through the optical coating, or antireflection layer, which lowers the amount of light lost through reflection and efficiently traps the light that hits the solar cell by stimulating its transfer to the energy-conversion layers below. An oxide of silicon, tantalum, or titanium is generally used to make the antireflection layer, which is frequently produced by spin coating or vacuum deposition on the cell surface.

The three energy-conversion layers beneath the antireflection layer are the top junction layer, the absorber layer, which functions as the device's core, and the back junction layer. Before reaching an external load and returning to the cell, the electric current must pass through two additional electrical contact layers, completing an electrical circuit. [6].

The electrical contact layer, which is frequently arranged in some grid pattern, is made up of a good conductor, such as a metal, on the cell face where light enters. Given that metal absorbs light, the grid lines are as thin and far apart as they may be without impairing the ability to collect the current produced by the cell. There are no boundaries that are so radically opposed in the back electrical contact layer. It just needs to operate as an electrical contact, therefore it must entirely cover the back surface of the cell structure. Because it must also be an excellent electrical conductor, the rear layer is always made of metal.

Layers

1. Light absorbing material

It is a semiconductor substance that serves as the main structural element of a solar cell and absorbs solar energy. And due to its abundance on Earth, silicon is the material most frequently used for solar cells, as was already said. Charge carriers are created as a result of light absorption and are subsequently transported to contacts. To achieve high solar energy conversion efficiency to electricity, some characteristics of this material, such as absorption, technological processing challenges, cost, and availability, should be addressed[7].

2. Solar trapping layers

These layers work to either enhance light transmission through the solar cell's front surface or lessen light egress once it is within the cell by reducing light reflections from the front side of the cell.

Antireflection coatings (the most popular of which is silicon nitride (SiN), which gives solar cells a blueish appearance). Surface tackiness (chemically etched or laser assisted surface structure, what resembles pyramids). Oftentimes, the front surface will blend these two techniques. The back of the cell has only texturing.

3. p-n junction

For the electrons produced by solar light to move to contacts, a potential voltage that is built into a solar cell functions as a driving force. An unusual type of connection called a pn junction is used to generate this inherent potential. The electronics industry of today has come to depend on this kind of connection.

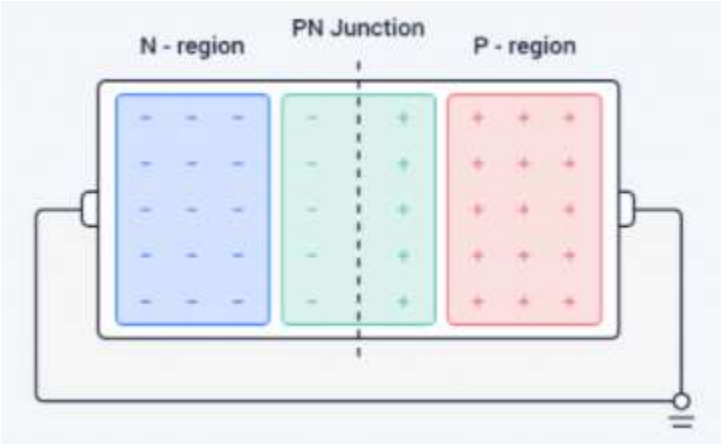


Figure 2 p-n junction of a solar cell

4. Charge collection terminals

A solar cell must have contacts in order to transmit charge to terminals, accumulate it, or be directly utilised with, instance, a household appliance. Different contact structure variations are employed in photovoltaics. This type of metallization consists of a metal grid (often silver paste) on the cell's front side and a full sheet of metal (typically aluminium) on the back. Since there will be less resistance and the charge carrier's path will be shorter, it makes sense that having more metal would be preferable. Contrarily, metals absorb all of the light and cause the cells to function inefficiently by casting shadows over them[8].

Operation:

In the absorber layer of a solar cell, excited electrons go from their lower-energy "ground state," where they are bound to specific atoms in the solid, to their higher-energy "excited state," where they are free to move around the solid. Since these "free" electrons are moving randomly in the absence of the junction-forming layers, there cannot be directed direct current. The photovoltaic effect, however, is produced by an intrinsic electric field brought about by the insertion of junction-forming layers. The electric field actually gives the electrons a collective motion so they can pass through the electrical contact layers and enter an external circuit where they might conduct useful work.

The internal electric field and the conductivity of the electric current depend on the materials of the two junction-forming layers, which must be different from those of the absorber. Therefore, these may be two different kinds of semiconductors (or one semiconductor with two different types of conduction), or they could be a semiconductor and a metal. Diodes and transistors used in microelectronics and solid-state electronics, as well as the several layers of solar cells, are made of similar materials[9].

In some ways, the photovoltaic process is similar to photosynthesis, the process by which plants convert light energy into chemical energy. Solar cells can't, of course, make electricity in the absence of light, therefore they frequently store some of the energy they produce in the presence of light for later use. One usual technique of storing this electrical energy is through the charging of electrochemical storage batteries. A striking similarity exists between this process of turning light energy into excited electron energy and finally into chemical energy that is stored.

Significance of material:

Crystalline Silicon:

The two different varieties of crystalline silicon are monocrystalline silicon (mono-Si) and polycrystalline silicon (poly-Si). Silicon, which is monocrystalline, is homogeneous. It is composed of one continuous crystal and has no grain boundaries. The silicon atom orientation and lattice parameter are both consistent throughout the entire substance. Contrarily, polycrystalline silicon is composed of several small silicon crystals[10].

Amorphous Silicon:

Amorphous Silicon (a-Si) is a non-crystalline.

Photovoltaic properties comparison

Photovoltaic properties	Amorphous Silicon	Crystalline Silicon
Optical-electrical conversion efficiency	14.0%	22.3%-26.1%
Bandgap	1.75eV	1.1eV
Area required/kW(m^2)	15	7-8
Sufficient thickness	1-2μm	100 μm

Optical-electrical conversion efficiency

The conversion efficiency of a solar cell is a critical consideration when assessing its performance. Light power is measured using the same metric: the higher the electrical output, the better. Amorphous silicon solar cells continue to be somewhat efficient, as shown in Table 1, even as crystalline silicon solar cells become more efficient. It is obvious that

c-Si solar cells are more effective at converting sunlight into electricity than a-Si solar cells (H Kang 2021). The reason why has a structural basis. In order for a solar cell to function properly, photons must be absorbed in order to create charge carriers, which must then be transported and collected in order to generate electricity. Hydrogenation reduces the number of dangling bonds making amorphous silicon more accessible, but the defect density of 10^{16}/cm3 compared to crystalline silicon is still dramatic (Hart, Paul R.1987).

Absorption Coefficient

The capacity to absorb light is a crucial element influencing solar cell performance. Amorphous silicon displays clear benefits in terms of the absorption coefficient. The study of structure in language is still very significant. Because of the clearly defined crystalline structure, crystalline silicon has internal homogeneities that are represented by clearly specified bond lengths and bond angles. The internal scattering of light has little impact. Contrary to popular belief, amorphous structure indicates that the bonds within the substance are unstable and disrupted, which can cause internal inhomogeneities. Internal light scattering is facilitated by these imperfections within. Amorphous silicon solar cells can effectively absorb light as a result (Staebler, D.L., and C. R. Wronski. 1977).

At least two advantages may result from high absorption coefficients. Amorphous silicon requires less material than crystalline silicon, assuming the same amount of light is absorbed. In a-Si solar cells, the total thickness of the absorbing layers is approximately 1 to 2 m, whereas in c-Si solar cells, it is approximately 100 m. Compared to cells based on photosynthesis, conventional cells require a hundred times greater amount of material to absorb sunlight. Furthermore, a larger absorption coefficient results in a better light capture in diffuse and low light settings(H Kang 2021). Therefore, depending on this factor, amorphous silicon solar cell is a better option for indoor usage than the crystalline silicon solar cell.

Electrical stability

Considering electrical stability is also another important factor. There is a price for this product's hydrogenation. The Staebler-Wronski Effect (SWE), which is exclusive to amorphous silicon solar cells, is one type of light-induced deterioration. In general, as illumination increases, the advantages of hydrogenation on amorphous silicon decrease. The efficiency of silicon solar cells decreased by 10-15% within the first 100 hours of light exposure, as indicated by a decline in photoconductivity and dark conductivity. The fact that the Staebler-Wronski effect is metastable is particularly intriguing. Cell performance is largely steady after the first 100 hours of exposure to light(Staebler, D.L., and C. R. Wronski. 1977).

The Staebler-Wronski Effect is still mostly unexplored. The hydrogen bond swapping theory is probably accurate. Near a weak Si-Si bond, photogenerated carriers may recombine and release energy sufficient to break the Si-Si bond. The dangling link is subsequently left when the nearby H atom establishes a new bond with one of the silicon atoms. By trapping electron-hole pairs, these dangling bonds can lower the current across them (Stutzmann, M. 1986). When compared to the most recent experimental findings, the answer gets more complicated. Although the relationship between hydrogenation and light-induced degradation is obvious, it is still unknown where the light-induced flaws exactly are. Because integrated hydrogen atoms and dangling bonds do not have a spatial link, comprehending Stabler-Wronski phenomena is extremely difficult.

Even Nevertheless, a lot of research has suggested a number of actions that could be taken to slow the degradation. By lowering the dilution during film growth, it is possible to obtain optimal hydrogenated amorphous silicon (Fritzsche, Hellmut. 2001). The Staebler-Wronski effect would rarely be diminished without compromising the quality of the

material since hydrogenation is employed to lower flaw density. SWE can be better understood by using cutting-edge multi-junction techniques or by further reducing the thickness of a-Si:H solar cells.

Temperature Stability

In sunlight, solar cells operate. Heat absorption is therefore inevitable. Under bright sunlight, the temperature of the cell can often reach over 70°C. When the temperature increases by 1°C over 25°C, crystalline silicon solar cells lose power by 0.4% to 0.5%. A 1°C increase in temperature, on the other hand, results in a 0.2–0.25% reduction in the power of amorphous silicon solar cells. Comparatively to crystalline silicon solar cells, amorphous silicon solar cells are often temperature-insensitive[11].

Applications

Conditions where Crystalline Silicon is more appropriate:

- The available space for installation is constrained, and it might not be perfect for the project. C-Si solar cells have a high conversion efficiency, which reduces the amount of space needed to produce 1 kW of power by half compared to a-Si solar cells.
- Price is impacted. At the moment, c-Si solar cells are about half as expensive per watt as a-Si solar cells[12].

Conditions where amorphous Silicon is more appropriate:

- In environments with artificial (indoor) illumination. With a higher absorption coefficient and spectral range resembling artificial light sources, the a-Si solar cell is a superior option for usage in artificial light applications.
- There are weight-related considerations. As was already mentioned, a-Si solar cells are more thinly layered than other solar cells. As a result, a-Si solar cells have a lower weight density—only one-fourth that of their crystalline counterparts. Consequently, lightweight solar cells do not add undue burden to buildings.
- Devices used in micro-power applications are frequently compact and have constrained functions. As a result, they are employed in certain applications like sensor networks and smart home technologies. For instance, a-Si solar cells are typically used in solar calculators due to their ability to be mass produced.
- This project's design aesthetics are crucial. A-Si solar cells' adaptability enables smooth device integration.
- In situations with high temperatures. The performance of solar cells as a whole is improved because a-Si solar cells are less susceptible to temperature than c-Si solar cells..
- Protecting the environment is important. One of the most environmentally benign uses of photovoltaic technology is a-Si solar cells because they don't contain any harmful heavy metals[12].

Photovoltaic Effect

In general, all solar cells operate on the same principle, namely the photovoltaic effect. The photovoltaic effect, in general, is the development of a potential difference at the junction of two distinct materials in response to visible or other light.

The basic processes behind the photovoltaic effect are:

- Charge carrier generation as a result of photon absorption by materials creating a junction.
- The subsequent separation of the charge carriers created by photons in the junction.
- The accumulation of photo-generated charge carriers at the junction's terminals.

In a solar cell, electron-hole pairs are formed by the effective photon absorption of the absorber layer, which is part of the structure of the solar cell. On both sides of the absorber, so-called "semi-permeable membranes" are attached to separate the electrons and holes produced by the photons(P. Wurfel. 2005). Only allowing one type of charge carrier to pass through is the most important requirement for semi-permeable membranes. For a solar cell to function effectively, it must be possible for electrons and holes made in the absorber layer to reach the membranes. Since the absorber thickness must be higher than the charge carrier diffusion lengths, this is necessary. Having a high conductivity for electrons and a low conductivity for holes creates a membrane that allows electrons to pass through but prevents holes from doing the same[13].

The p-n junction

Currently, crystalline silicon is used to create the most popular version of the solar cell construction mentioned above (c-Si). With an acceptor concentration of 1016 cm-3, a moderately doped p-type c-Si is used as the absorber. A thin, highly-doped n-type layer less than 1 m thick is formed on the top side of the absorber and serves as the electron membrane. On the back of the absorber, a heavily doped ptype serves as the hole membrane. The c-Si p-type absorber and the heavily doped n- and p-type membranes come together to form an internal electric field at these interfaces. P-n junctions are these regions, which are crucial for solar cells. A solar cell's internal electric field facilitates the separation of electron-hole pairs produced by photolysis. Charge carriers do not contribute to energy conversion when they are annihilated in a procedure known as recombination when they are not separated from one another in a reasonably short period of time. The simplest method to separate charge carriers is to place them in an electric field. In an electric field, carriers with opposing charges float away from one another in opposite directions and can connect to the electrodes of a solar cell. The term "electrodes" refers to the metal contacts that are affixed to the membranes (P. Wurfel. 2005).

A p-n junction that is constructed of the same semiconductor material, such as c-Si, is known as a p-n homojunction. Different junction types produce an internal electric field within the junction. A p-n junction made of two semiconductors with different chemical compositions is known as a p-n heterojunction. By adding an intrinsic, I layer between the p-type and n-type layers, p-i-n junctions expand the area of the internal electric field. The i-layer functions as a capacitor, extending the p-n junction's electric field across itself. Another kind of junction that joins a metal and a semiconductor is called an MS junction. The Schottky barrier that is produced at the metal-semiconductor contact is a typical illustration of an MS junction.

Formation of space-charge region

The p-type and n-type semiconductor components in the figure below, together with the associated band diagrams, are schematically isolated. In both isolated regions, the neutral charge is preserved. In an n-type semiconductor, there are many negatively charged free electrons, which is balanced by positively charged ionised donor atoms. The bulk of carriers in a ptype semiconductor are holes, and the negatively charged ionised acceptor atoms counteract their positive charge.

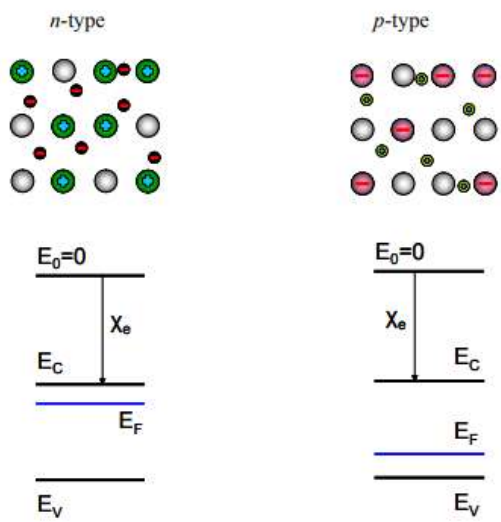


Figure 3 Band diagram and schematic of an isolated p-type and n-type semiconductor

Diffusion is induced when an n- and p-type semiconductor are joined because there is a significant difference in the electron concentrations between them. The metallurgical junction is the conduit for the electron flow from the n-type material into the p-type material. An analogous hole diffusion current from p- to n-type material is induced by a change in hole concentration. This mechanism of diffusion almost entirely exhausts the area where mobile charge carriers meet metallurgical processes. Space charges created by the charge of the ionised donor and acceptor atoms are no longer balanced by mobile charges as a result of the gradual depletion of charge carriers. The depletion region is where this space charge is located[14].

p-n junction in dark (External applied voltage)

A p-n junction's electrostatic potential across the space-charge region changes to (0 - Va) when an external voltage, Va, is applied. The potential difference between the n- and p-type regions also changes. The potential difference across the p-n junction widens as the applied external voltage is negative relative to the potential of the p-type region. In the presence of reverse bias voltage, this is referred to as a p-n junction. There is a larger space-charge zone when reverse-bias voltage is applied because the potential barrier across the junction rises.

Under reverse bias, the p-n junction's band diagram is shown in Figure 2. The quasi-Fermi energy for electrons (EFC) and the quasi-Fermi energy for holes (EFV), respectively, describe the electron and hole concentrations at the p-n junction when it is not in equilibrium due to external voltage. The potential difference across the p-n junction is reduced when the external voltage is applied and is positive relative to the potential of the p-type region[15].

Under forwardbias voltage, this is referred to as a p-n junction. The band diagram of the forward-biased p-n junction is shown in Figure 2b. The space charge area gets smaller and the potential barrier across the junction gets lower as the junction is forward biassed. Diffusion- and drift-causing forces, concentration gradient and electric field, are not in balance. Near the edges of the space-charge region, the concentration of minority carriers is higher than in equilibrium when the electrostatic potential barrier is lowered. Minority-carrier injection is the name of this method.

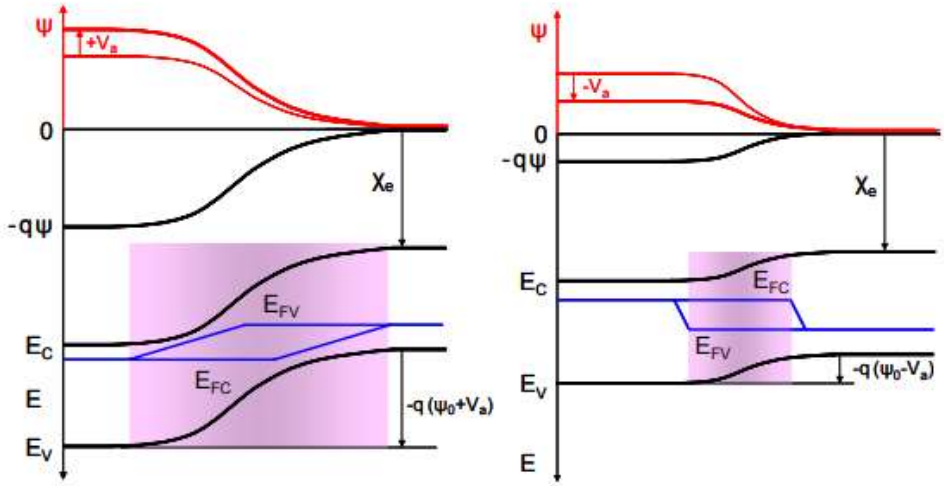


Figure 4 Energy band diagram and potential of p-n junction under a) reverse bias b) forward bias

I-V characteristics:

The dark IV curve of a solar cell diode and the current produced by light make up the solar cell's IV curve. 1 The IV curve is shifted by the light so that it descends into the fourth quadrant, which is where the diode's power may be recovered. The diode rule becomes more applicable when a cell is illuminated because it raises the normal "dark" currents in the diode.

$$I = I_0 \left[\exp\left(\frac{qV}{nkT}\right) - 1\right] - I_L$$

The equation in the first quadrant becomes,

$$I = I_L - I_0 \left[\exp\left(\frac{qV}{nkT}\right) - 1\right]$$

In most circumstances, it is possible to ignore the -1 term in the equation above. The exponential term is often > 1, with the exception of for voltages below 100 mV. Furthermore, the -1 term is superfluous at low voltages because the current that produces light, IL, exceeds the I0 (...) term.

$$I = I_L - I_0 \left[\exp\left(\frac{qV}{nkT}\right)\right]$$

The IV curve is produced by plotting the aforementioned equation, and the critical places are highlighted and further discussed on the pages that follow. The maximum point of the power curve, shown as PMP, is where the solar cell should be operated to generate the greatest electricity. It occurs when the voltage is VMP and the current is IMP, it is also referred to as PMAX or maximum power point (MPP)[16].

The current-voltage (I-V) characteristic of an ideal p-n junction is illustrated schematically in figure below,

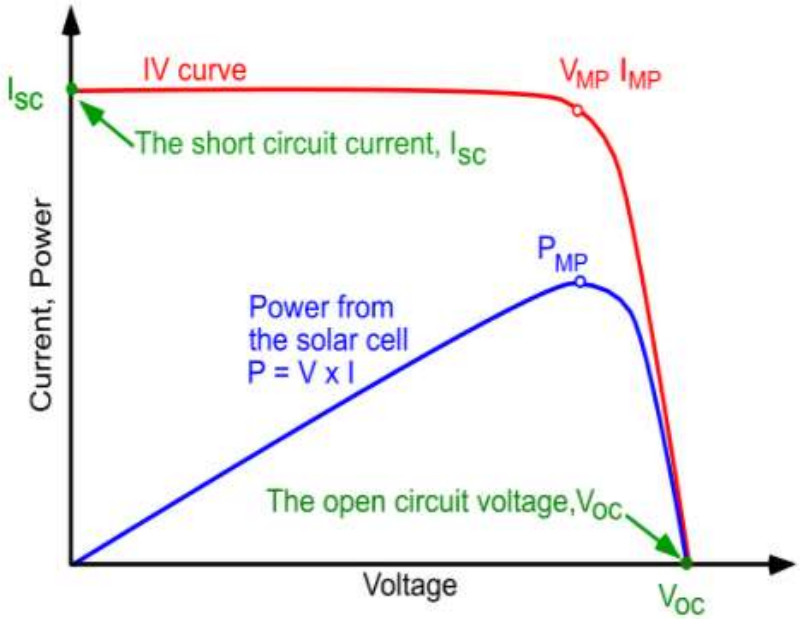


Figure 5 I-V characteristic of a p-n junction (Solar cell)

C-V characteristics

It is possible to estimate the charge carrier density and the driving level capacitance profile using measurements of capacitance voltage, which are some of the fundamental properties of solar cells. Simple electronics is used in the quasi-static capacitance-voltage (C-V) technique to gradually increase the reverse-bias voltage on the solar cell in the dark in order to assess the influence of junction capacitance on the bias voltage[17].

p-n junction under illumination

When a p-n junction is illuminated, more electron-hole pairs are produced in the semiconductor. Electrons in the p-type region and holes in the n-type region, which are minor carriers, become increasingly concentrated. Minority carriers flow through the depletion zone and into the qausineutral region as a result of the increase in minority carrier concentration. While holes move from the n-type region into the p-type zone, electrons move from the p-type zone into the n-type zone. The photo-generation current is produced by the movement of photo-generated carriers. Using a band diagram, Figure 1 shows the situation of an illuminated p-n junction in an open circuit[16].

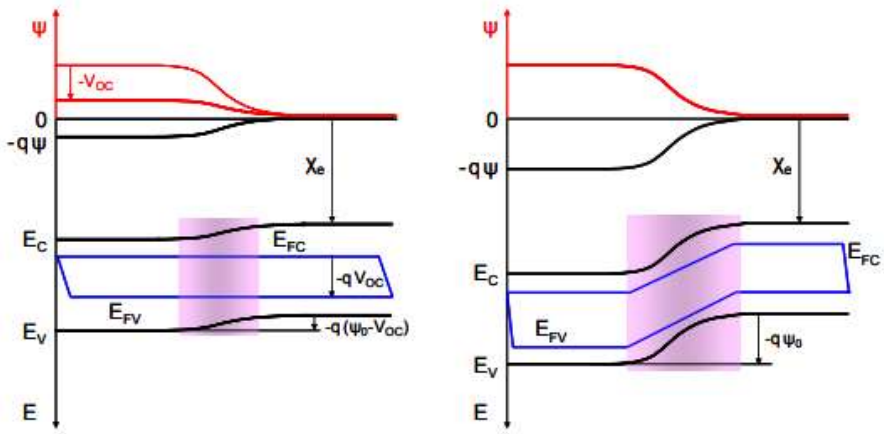


Figure 6 Energy band diagram and electrostatic-potential of an illuminated p-n junction a) open-circuit and b) short-circuit conditions.

Voc lowers the electrostatic-potential barrier over the connection. Voc stands for the voltage in open circuit. The concentrations of electrons and holes in non-equilibrium conditions are characterised by the quasi-Fermi energy levels. Figure 2 demonstrates that the n-type zone has a higher electrochemical potential for electrons (EFC) than the p-type region (q Voc). The photo-generated current also flows via the external circuit when the n-type and p-type sections are shorted under illumination Figure 2 depicts this condition. In the case of a short circuit, the electrostatic potential barrier is unaffected, but a significant variation in the quasi-Fermi levels inside the depletion zone suggests that current is flowing inside the semiconductor. Only a portion of the photo-generated current travels through the external circuit when a load is connected between the electrodes of the lighted p-n junction. The electrochemical potential difference between the n-type and p-type regions will be lessened by a voltage drop across the load. Eqs. , which describes how the junction behaves when a voltage is applied, is supplied to describe the net current of the lit p-n junction,

$$I(Va) = Irec(Va) - Igen(Va) - Iph = I0\left[\exp\left(\frac{qVa}{kT}\right) - 1\right] - Iph$$

Figure 3 shows the dark and illuminated I-V characteristics of the p-n junction. The superposition concept is depicted in Figure. The photo-generated current Iph causes the illuminated I-V characteristic of the p-n junction to go lower than the dark I-V characteristic[16].

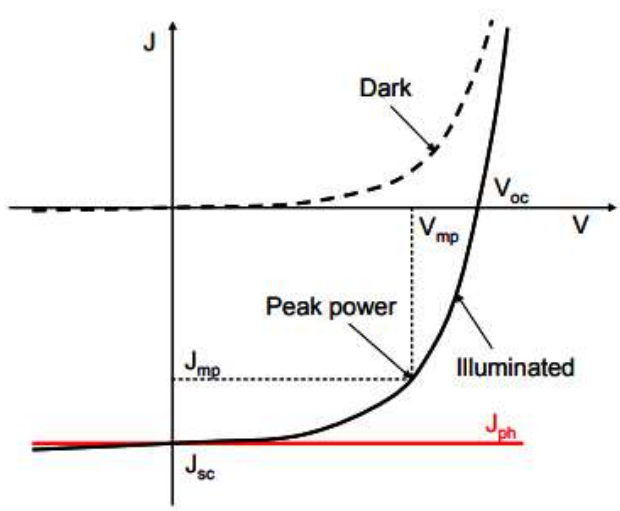


Figure 7 I-V characteristics of a p-n junction in dark and illumination

Solar cell parameters from dark and light conditions:

The three main factors used to describe the performance of solar cells are the short-circuit current (Isc), open-circuit voltage (Voc), and fill factor (FF). Utilizing the lighted J-V characteristic depicted in Figure 4.10, these values are computed. The conversion efficiency is determined by these factors. The current that flows through the external circuit when the electrodes of a solar cell are shorted is known as the short-circuit current. In a perfect world, the photogenerated current, Iph, and Isc are equal. The open-circuit voltage is the voltage at which the external circuit is not conducting any current. In this case, the dark current balances out the photo-current[17].

The open circuit voltage is dependent on the photo-generated current, which can be calculated by the equation assuming that the net current is zero,

$$Voc = \frac{kT}{q}\ln(\frac{Iph}{I0} + 1)$$

The short-circuit current and open-circuit voltage of a solar cell are their maximum values, respectively. However, at both of these operational locations, the solar cell's output is zero. The "fill factor," often known as "FF," along with Voc and Isc, is a parameter that affects a solar cell's maximum power output. The maximum power generated by a solar cell divided by the product of Voc and Isc is known as the FF:

$$FF = \frac{Imp * Vmp}{Isc * Voc}$$

Series and Shunt resistance

Three things contribute to series resistance in a solar cell: first, the flow of current between the emitter and base; second, the resistance at the point where a metal contact makes contact with silicon; and third, the resistance at the top and rear metal contacts. Although extremely high values can also reduce short-circuit current, series resistance's main impact is a reduction in fill factor.

Since there is no current flowing through the solar cell at open-circuit voltage, there is no effect of series resistance on the solar cell. But the IV curve close to the open-circuit voltage is significantly influenced by the series resistance. The series resistance of a solar cell can be calculated by calculating the slope of the IV curve at the open-circuit voltage point.

Instead of poor solar cell design, significant power losses brought on by shunt resistance, or RSH, are typically the result of manufacturing errors. Low shunt resistance results in power losses in solar cells by providing an alternative current route for the light-generated current. A diversion of this kind lowers the voltage generated by the solar cell by reducing the current flowing through the junction of the solar cell. The impact of a shunt resistance is particularly strong at low light levels since there is less light-generated current there. The impact of losing this current to the shunt is therefore larger. Additionally, a parallel resistance has a large impact at lower voltages where the solar cell's effective resistance is high.

Open circuit voltage (V_{oc})

The open-circuit voltage, or VOC, is the maximum voltage a solar cell can generate while there is no current passing through it. The solar cell's forward bias, which is brought on by the junction's bias with light-induced current, is represented by the open-circuit voltage.

It can be determined by the following equation,

$$V_{oc} = \frac{nkT}{q} \ln\left(\frac{I_L}{I_0} + 1\right)$$

One might conclude from a superficial look at the aforementioned equation that VOC concentrations rise linearly with temperature. However, this is untrue because I0 rapidly increases with temperature, primarily due to changes in the intrinsic carrier concentration.

Short circuit current (I_{sc})

The short-circuit current travels through the solar cell while there is no voltage across it (i.e., when the solar cell is short circuited). Short-circuit current, commonly spelt ISC, is created when light-generated carriers are generated and gathered. For a perfect solar cell, the short-circuit current and the current that generates light are equal at most minor resistive loss mechanisms. Short-circuit current is therefore the greatest current that can be extracted from a solar cell.

The short circuit current depends on the following factors:

- The area of solar cell
- The optical properties
- Spectrum of the incident light
- Number of photons.

Ideality Factor

The ideality factor of a diode is a gauge of how closely it adheres to the ideal diode equation. The formulation of the fundamental diode equation makes some assumptions about the cell. The ideality factor describes the second order factors that, in practice, lead to the diode deviating from the straightforward diode equation. The equation for the ideality factor can be given as,

$Y = a + mx,$

Where, m is the slope of the graph and can be given as,

$m = \frac{40}{n}.$

Efficiency

The ratio of the highest power generated to the incident power is used to calculate conversion efficiency. Solar cell conversion efficiency is now calculated using the irradiance value, Pin, of 1000 W/m2 of AM1.5 spectrum.

$$\eta = \frac{Pmax}{Pin} = \frac{Imp * Vmp}{Pin} = \frac{Isc * Voc * FF}{Pin}$$

Lifetime

The lifespan of minority carriers is very important in solar cells because to its impact on potential cell efficiency.

There are basically two primary categories of carrier lifetime:

1. Recombination lifetime: The disappearance of extra minority carriers as a result of recombination.
2. Generation lifetime: the typical time required to create an electron-hole pair in areas with a shortage of carriers, like the space charge region[18].

Impedance Spectroscopy

Impedance is a measurement of an electronic component, circuit, or system's resistance to alternating and/or direct electric current. Two distinct scalars combine to form the two-dimensional vector quantity known as impedance (one-dimensional)

Resistance is a measure of an object's resistance to the movement of electrons between its atoms. The resistance, which is expressed in positive real number ohms, reduces when atoms readily give up or accept electrons. Direct current (DC) and alternating current (AC) both have resistance (DC). Electrical conductors, or substances with low resistance, include things like copper, silver, and gold. High resistance materials include insulators and dielectrics, such as mica, glass, and polyethylene. A substance with a middle-range resistance is known as a semiconductor. Galium arsenide, germanium, and silicon are some examples.

Reactance, denoted by the symbol X, is the term used to represent how much energy an electrical component, circuit, or system retains and expels when the current and voltage vary throughout each AC cycle. The ohm is a made-up unit used to quantify reactance. AC causes it to be noticeable, but DC does not. Depending on whether the reactance is capacitive (denoted +jXL) or inductive (denoted +jXL), energy can be retained and released as either a magnetic field or an electric field when AC travels through the component (denoted -jXC). Reactance is often multiplied by the positive square root of -1, also known as the j operator, in order to describe Z as a complex number of the form R + jXL (when the net reactance is inductive) or R - jXC (when the net reactance is capacitive).

Impedance spectroscopy has been widely used to examine electrochemical systems and solid-state electronics in recent decades. However, due to the odd material properties and complex device topologies, performing impedance spectroscopy on developing photovoltaics brings new hurdles.

The capacity of a circuit component to resist electrical current passage. Resistance is defined by Ohm's law (Equation 1) as the proportion of voltage to current.

$$R = \frac{V}{I} \tag{1}$$

This equation is for linear circuits, an ideal resistor have a set of properties which are:

- It obeys ohm’s law for all voltage and current levels.
- The resistance value is independent of frequency.
- The AC current and voltage signals through a resistor are in phase with each other[19].

However, the behaviour of circuit components in the real world is much more complex. These reasons urge us to use impedance, a more common circuit characteristic, in place of the easy concept of resistance. Similar to resistance, impedance measures how well a circuit may hinder the flow of electrical current, however unlike resistance, impedance is not limited by the aforementioned simplifying features. Usually, an electrochemical cell is given an alternating current potential, and the current flowing through the cell is measured to calculate the electrochemical impedance. Typically, a

brief pulse of low excitation is used to assess electrochemical impedance. To ensure that the solar cell's response is pseudo-linear, a modest excitation is applied.

The impedance is represented as complex number,

$$Z(w) = \frac{V}{I} = Z_0 \exp(j\Phi) = Z_0(cos\Phi + jsin\Phi) \quad (2)$$

Equivalent circuit :

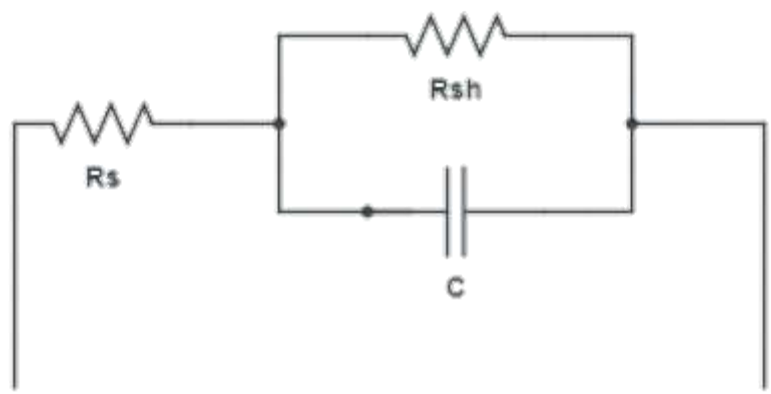


Figure 8 Equivalent circuit of a solar cell

The above figure represents the equivalent circuit of a solar cell (p-n junction), it consists of a resistor in series with a parallel capacitor and resistor connection.

Circuit elements:

Usually, fitting to a model of an equivalent electrical circuit is used to analyse IS data. The majority of the circuitry in the model consists of standard electrical parts including resistors, capacitors, and inductors. Therefore, it is quite beneficial to grasp the impedance of the fundamental circuit components. Table 1 lists the common circuit components, their impedance, and the equation for the relationship between their current and voltage.

Table 1 Impedance of electrical components

Component	V vs I	Impedance
Resistor	V = IR	Z = IR
Inductor	V = L di/dt	Z = jwL
Capacitor	I = C dV/dt	Z = 1/jwC

The impedance of a resistor has no imaginary component and is frequency independent. The current flowing through a resistor stays in phase with the voltage across the resistor when there is only a real impedance. An inductor's impedance rises as its frequency rises. Only a fake impedance component exists in inductors. As a result, the voltage and current via an inductor are phase-shifted by -90 degrees. In contrast to an inductor, a capacitor behaves differently in terms of impedance against frequency. As the frequency rises, a capacitor's impedance decreases. Capacitors also have an imaginary impedance component. In relation to the voltage, the phase of the current flowing through a capacitor is rotated by 90 degrees.

So, the impedance for this circuit can be given as:

$$Z = R_s + R_{sh}||X_c \quad (3)$$

$$Z = \frac{R_s+R_{sh}+w^2R_{sh}^2C^2R_s}{1+w^2R_{sh}^2C^2} - \frac{jwR_{sh}^2C}{1+w^2R_{sh}^2C^2} \quad (4)$$

The above equation is the final form of the impedance, the first part is the real part and the latter is the imaginary part, we can further separate the real part and the imaginary part of this equation and differentiate with respect to w, and equate it to zero, we can get the Z_{min} and w values.

$$w = \frac{1}{R_{sh}C} \text{ , w at } Z_{min} \quad (5)$$

And,

$$Z_{min} = \frac{R_{sh}}{2} \quad (6)$$

Data Analysis :

Impedance can be plotted as a Nyquist plot since it may be stated as a complex number (the real part on the x-axis, and the imaginary part on the y-axis). The shape of this plot is often a semicircle, signifying a charge transfer process. The size of the semicircle indicates the amount of charge transfer resistance present. However, if it is a diffusion process, this plot can also appear to be a straight line with a positive slope (called Warburg impedance). The circuit also plays a role in this plot (parallel, series, or combinations). Impedance can also be shown as a Bode plot in addition to Nyquist plots**.**

There are real and imaginary components to the formula for Z, according to the equation (w). Plotting the imaginary component on the Y-axis and the actual component on the chart's X-axis creates a "Nyquist Plot" (figure 1). The Nyquist Plot's points represent the impedance for each frequency, and the Y-axis is oriented negatively in this diagram. According to Figure 1's annotations, lower frequency data are presented on the right side of the plot and higher frequency data on the left. The angle between the x-axis and this vector on a Nyquist plot is known as the "phase angle," and impedance can be represented as a vector of length |Z|[20].

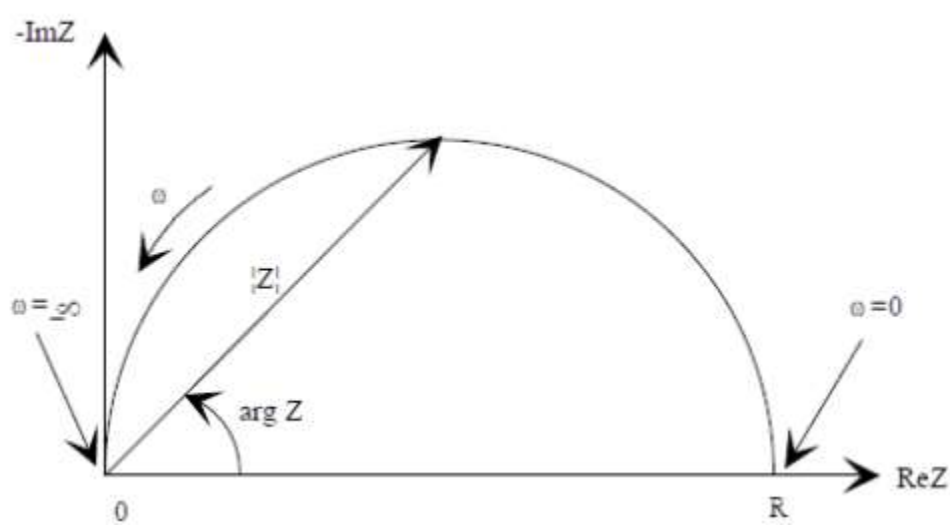


Figure 9 Nyquist plot with Impedance vector

Nyquist Plots contain one significant flaw. When examining any data point on the graph, it is impossible to determine the frequency utilised to record that point. The semi-circle is the characteristics of a good cell behavior, the better the semi-circle the better is the efficiency of the solar cell, this way by analyzing the impedance characteristics of the cell, the efficiency of the cell can be determined.

The Nyquist plot for an ideal solar cell circuit mentioned above is always a semi-circle, the series resistance can be calculated by reading the real axis at the high resistance intercept. It is near the origin of the plot, the real axis value ath the low frequency is the sum of the series and shunt resistance, the diameter of the semicircle, therefore, is the shunt resistance of the cell[21].

The Bode Plot is another frequent approach to illustrate impedance. On the X-axis, the impedance is displayed with log frequency; on the Y-axis, the absolute values of the impedance (|Z|=Z0) and the phase-shift are plotted. Bode plot representation can be seen in the figure below, The Bode Plot displays frequency data, whereas the Nyquist Plot does not.

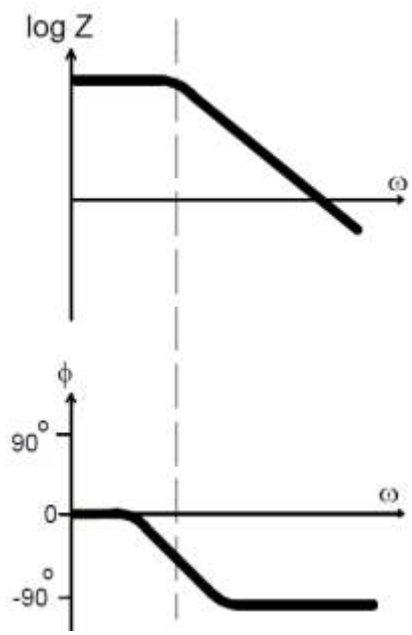


Figure 10 Bode plot representation with One time constant

Categorization

We have more than enough potential energy from the Sun each day to meet all of our needs for power generation. But as was already established, only around 15% of the light that reaches solar cells is actually captured by them. Naturally, a solar panel or array can capture more energy the larger it is.

We shall examine the most prevalent variety of crystalline silicon solar cells because their efficiencies differ from those of monocrystalline, polycrystalline, and thin film solar cells.

3-4.5 watts can be produced by a single solar cell, which is about the size of a compact disc. You can produce 100–300 watts by combining 40 of these cells into a module of a typical size. A solar panel made of several of these modules may produce several kilowatts of power, which should be sufficient to fulfil the peak energy demands of most households. According to calculations, it would take 22,000 solar panels spread across 30 acres to create 4.2 megawatts of power, which is enough to power 1,200 homes[22].

To put it into perspective, 500 to 1000 solar roof installations would be necessary to match the power produced by a large wind turbine (2 to 3 megawatts), and approximately one million solar roof installations would be necessary to match the output of a sizable coal or nuclear facility (which are rated in gigawatts)

On the basis of the impedance spectroscopy, we can calculate the series and shunt resistance from the graph, as well how good the diode behaves, the better the semicircle the better the behavior and the efficiency of the solar cell.

Methodology:

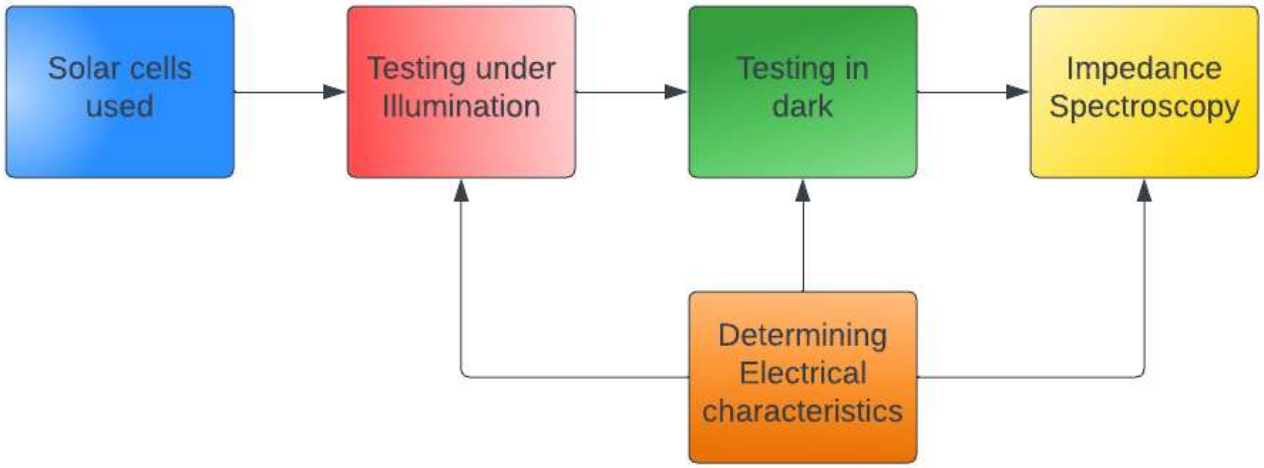


Figure 11 Methodology procedure

Solar cell used:

For this experiment, to collect different sets of data, three solar cells were used, all three made up of different material, namely,

- Monocrystalline silicon solar cell
- Polycrystalline silicon solar cell
- Amorphous solar cell

Mono-crystalline solar cell: This type silicon cells are frequently produced from a single, highly pure crystal ingot. The mono-crystalline cell used for testing is circular in shape, and consists of 4 cells, with a combined dimension of 2cm x 1.5 cm.

Figure 12 Mono-crystalline solar cell

Poly-crystalline solar cell: These cells are created by cooling and solidifying molten silicon to create cast square ingots for manufacturing. The cell used for the experiment has 4 cells, with a combined dimension of 4cm x 4cm.

Figure 13 Poly-crystalline solar cell

Amorphous solar cell: They are the non-crystalline, allotropic form of silicon used in semiconductors, the amorphous cell used for this experiment has a dimension of 5 cm x 4.6 cm.

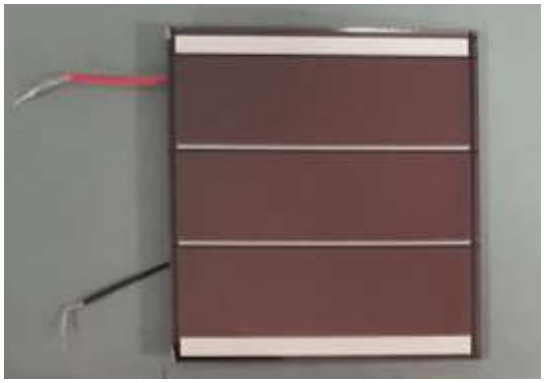


Figure 14 Amorphous solar cell

Testing Under Illumination:

For determining the cell parameters under illumination, a wavelabs SINUS-70, instrument was used as shown in figure 15, the cell is placed under the instrument, and then the chamber is closed, then with the help of the software beam of light is incident on the cells.

The software will collect all the data during this duration of time, for this experiment, we collected the data from 0V to 3V, the software will collect the data in steps from 0 volt to 3 volts, the data collected is current at each step along with the power. This software is also programmed to collect the short circuit current (Isc), Open circuit voltage (Voc), the maximum power point voltage, the maximum power point current, fill factor, the series and shunt resistance.

Figure 15 Instrument for testing under illumination

From the voltage and current data obtained, an I-V graph is plotted, which will be discussed further in results and discussion, usually the input power for calculations is taken as 1 kW/m^2, but for this experiment the device we used has the input power of 833 W/m^2, between the wavelength of 350-1200nm (Illumination Intensity used for this experiment).

Testing in dark (External applied voltage):

The testing of the p-n junction under dark is done with the help of the faraday's cage (as shown in figure 16), the cell is kept under observation, with the leads connected to the agilent voltage source, in which the cell is provided with an external voltage, while the cell is inside the cage and cage is closed, to observe the cell characteristics in complete dark environment.

Figure 16 Faraday's cage

After the setup is completed, through voltage source, we applied a voltage from 0V to 2V and again from 2V to 0V, and from 0 to -2V and back again to 0V.Make sure there is no short-circuit, or else the I-V characteristics will be wrong, By this procedure, we collected the voltage and current data for the cell. I-V characteristics shows the capability of a cell to convert sunlight into electricity.

Another test which was performed on the cell is C-V characteristics, this test is conducted with the same procedure, this test is carried out to check the junction of the diode in reverse bias, is it showing working characteristics or not, with this the barrier height of the cell is calculated, which can be determined by plotting $1/C^2$ vs V plot. The software used for this experiment is shown in figure below.

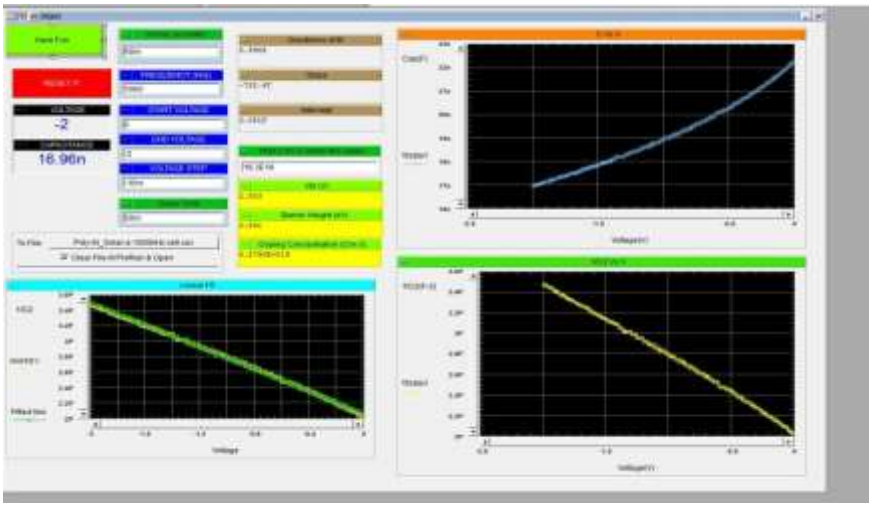


Figure 17 C-V measurement software

Once the data is collected, we need to plot the $1/C^2$ vs V graph, and do the linear fitting of the graph, from which we will get the slope, which is the barrier height of that junction.

Impedance Spectroscopy:

The task for this experiment is to analyze the behavior and merit of the cell through impedance spectroscopy, after running the tests under illumination and in dark, next step was to collect the impedance data of the cell, for this the equipment used is shown in figure 18, the cells were placed under observation, with the leads connected to the impedance analyzer, which gives the impedance data for the cell.

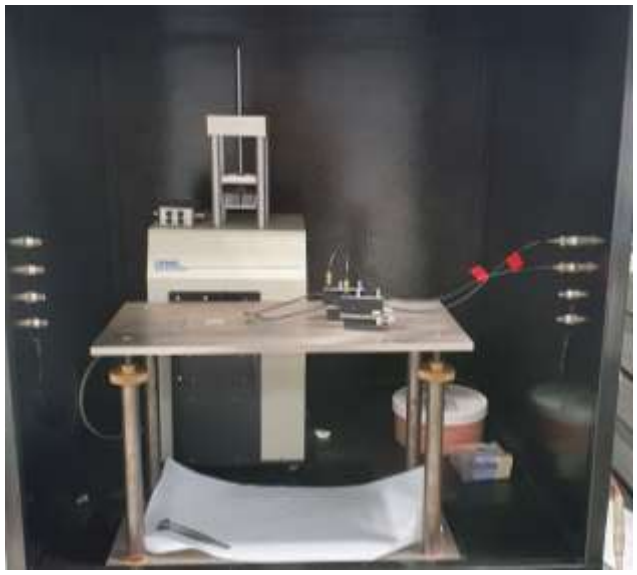


Figure 18 Testing equipment for Impedance analysis

The software used was Keysight Agilent VEE pro, it is used for the interfacing with the equipment, it is programmed to collect real and imaginary impedance data for the given range of frequency. A synopsis of the program is shown in the figure 19.

Figure 19 Program for Agilent VEE pro

For this part, we need to set the frequency range prior to running the program, for this experiment, we collected the data from 20Hz to 1MHz, with 4 different step size, for the initial 99Hz, the step size is 1, from 100Hz to 999Hz the step size is 5, for the next, till 1000Hz the step size is 500 and for the remaining till 1MHz the step size is 5000Hz, for each of

the frequency value the corresponding value of real and imaginary impedance is been collected through the impedance analyzer.

Next is, the software creates a set of data for all value of frequency from 20Hz to 1Mz and plots the graph for the same, frequency vs Z' (real), frequency vs Z"(imaginary) and Z'(real) vs Z"(imaginary).

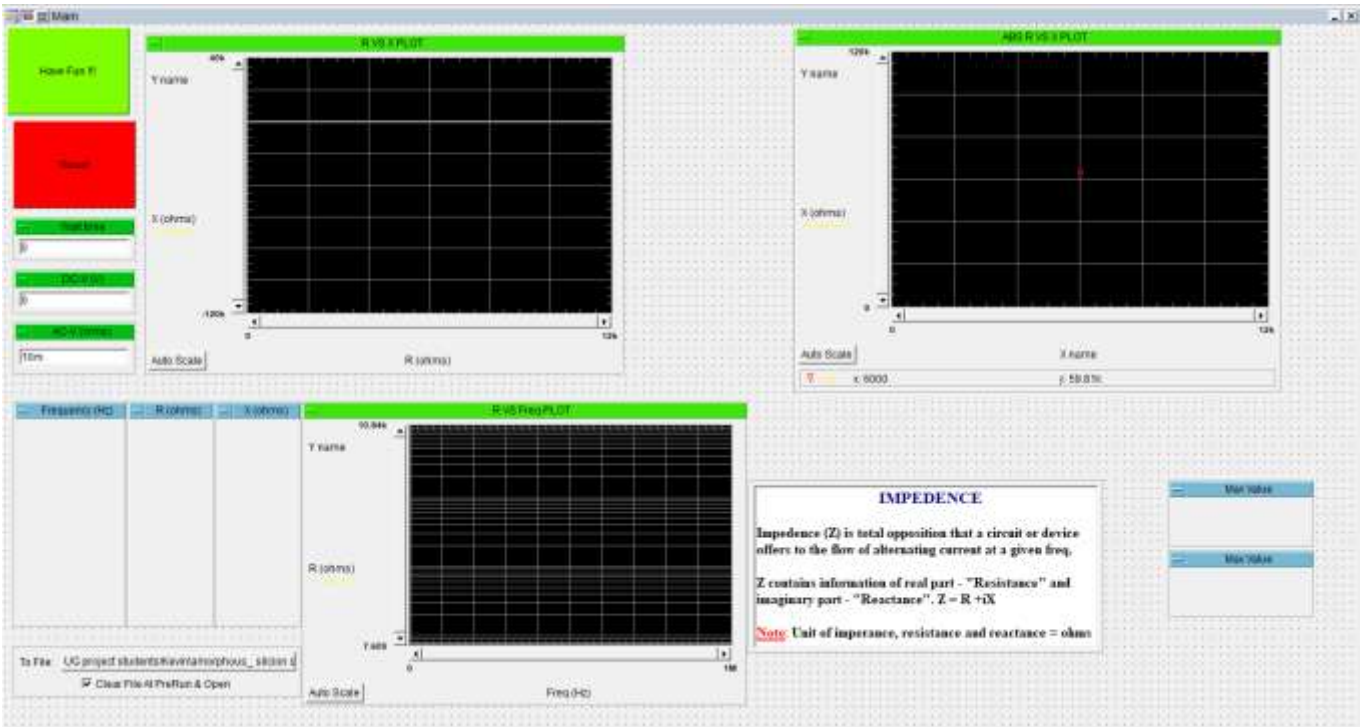


Figure 20 Outlook of the Agilent VEE pro

Results and Discussion

The data collected through the software in different conditions, under illumination, in dark and impedance spectroscopy will be analyzed and discussed in this part of the report.

Solar cell under Illumination characteristics:

The experiment is carried out for three different cells, all the measurements were carried out at T=293.15 K, starting with,

1. Poly-crystalline solar cell

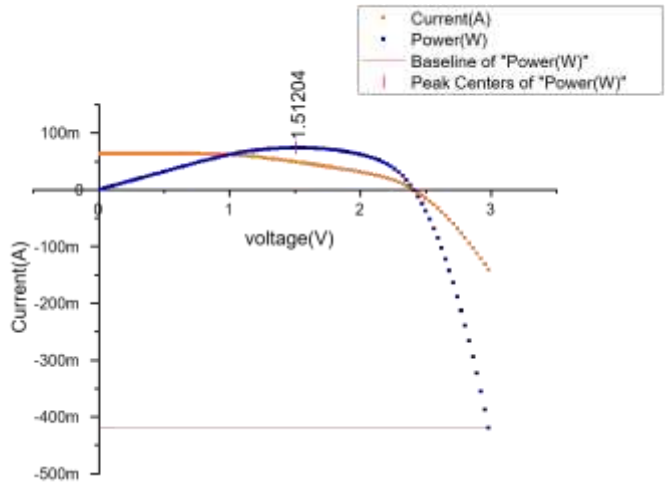


Figure 21 I-V measurements (in orange) and P-V measurement(in blue) under light for poly-crystalline cell

The graph above shows the current-voltage and power-voltage behavior of the cell under light conditions, as we can observe from the graph, the orange graph show the I-V behavior, as the voltage increases the current is decreasing, as we can see that the graph is plotted in the fourth quadrant, as the current is negative, which means the current is supplied by the cell and not drawn by the cell.

The PV curve is plotted to analyze the value of the current and voltage at the maximum power, the point at which the power curve intersect the X-axis is the Voc and at that point the value of current is Isc, the point at which the power is maximum, the voltage at that point is Vmp, and the current at that point is Imp.

There are various parameters which are calculated from this graph, Isc, Voc, the maximum current and voltage for calculating maximum power, which is required for calculating the fill factor, and the fill factor is required for the efficiency calculation.

from graph, we got Isc = 0.0646 A, Voc = 2.41 V and Pmax = 74.93 mW, hence, we can calculate the fill factor,

Fill factor = $\frac{Pmax}{Voc*Isc} = 74.93 * \frac{10^{-3}}{0.0646*2.41}$ = 0.481

The input power for the device we used is 833 W/m^2, and the dimensions of the poly-crystalline cell are 0.04m x 0.04m, hence, area = 0.0016 m^2,

Therefore, Pin = 833*0.0016 = 1.3328 W,

So, the efficiency = $\frac{V_{oc}*I_{sc}*FF}{P_{in}} = \frac{0.0646*2.41*0.481}{1.3328} = 5.6186\%$

Similarly, we calculated the shunt and the series resistance from the graph and other factors, which are shown in the table below and compared them with the values obtained from the software,

Table 2 Poly-crystalline cell under illumination parameters

Parameters	Calculated values	Software values
Isc (A)	0.0646	0.06369
Voc (V)	2.41	2.4
Imp (mA)	48.0	48.8
Vmp (V)	1.45	1.5
Pmax (mW)	69.6	73.89

Rshunt (kohms)	1.3	1.31
Rseries (ohms)	13.3	13.47
Fill factor	0.481	0.481
Efficiency	5.6186 %	

2. Mono-crystalline solar cell

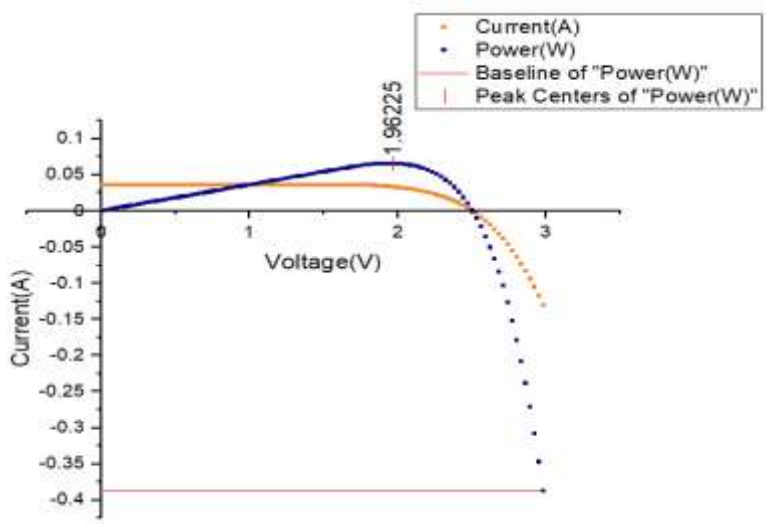


Table 3 I-V measurements (in orange) and P-V measurement(in blue) under light for mono-crystalline cell

From the I-V graph of mono-crystalline cell, we can observe the maximum power point which is at 1.962 V, which is the Vmp, and the current corresponding to that point is 0.038 A, which is Imp. For the mono-crystalline cell, the Voc is calculated to be 2.5 V from the graph and the short circuit current Isc is 0.0359 A from the above graph.

We have the Imp and Vmp, so we get the value of Pmp, With this data we can calculate the fill factor,

$$FF = \frac{P_{max}}{V_{oc} * I_{sc}} = \frac{0.0653}{2.5 * 0.0359} = 0.727$$

The input power for this cell will be 833*the area of the cell, here the dimensions of the mono-crystalline cell are 0.02m x 0.015m, hence, the area will be 0.003 m^2.

So, the input power for this cell will be 2.5W,

The efficiency of this cell will be,

$$\eta = \frac{V_{oc} * I_{sc} * FF}{P_{in}} = \frac{2.5 * 0.0359 * 0.727}{2.5} = 2.6\%$$

Other parameters calculated from the graph and software are listed below in the table,

Table 4 Mono-crystalline cell under illumination parameters

Parameters	Calculated values	Software values
Isc (A)	0.0359	0.0362
Voc (V)	2.5	2.5
Imp (mA)	33.2	33.48
Vmp (V)	1.962	1.96
Pmp (mW)	65.3	65.7
Rshunt(kohms)	4.72	4.83
Rseries(ohms)	1.9	1.93
Fill factor	0.727	0.724
Efficiency	2.6%	

3. Amorphous Silicon solar cell

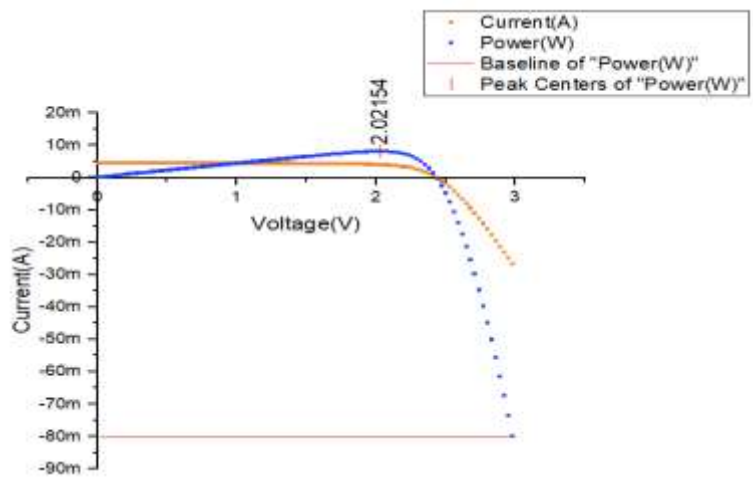


Figure 22 I-V measurements (in orange) and P-V measurement (in blue) under light for amorphous silicon cell

The IV curve (in orange) for the amorphous silicon cell is shown in figure 22, comparing with the mono-crystalline and poly-crystalline, we can observe that the curve is very close to the X-axis, which indicates that the IV characteristics of amorphous cell shows a poor behavior. Here, the voltage at the maximum power is 2.02V, which is the Vmp, and the current corresponding to it is 4 mA.

The open circuit voltage observed from the graph is 2.435 V, and the corresponding short circuit current is 4.49 mA, hence we can get the fill factor, the Pmax we got is 8.07 mW,

Fill factor = 0.739,

The input power for this cell will be 833*area of the cell, the dimensions of the amorphous cell is 0.05m x 0.046m, so the area of the cell will be 0.0023 m^2. So, the input power will be 1.9159 W.

The efficiency of the cell will be,

$$Efficiency = \frac{2.43 * 0.00449 * 0.739}{1.9159} = 0.42\%$$

Other parameters for this cell are listed in the table below, compared to the software values as well,

Table 5 Amorphous silicon cell under illumination parameters

Parameters	Calculated values	Software values
Isc (mA)	4.49	4.74
Voc (V)	2.435	2.43
Imp (mA)	4	3.99
Vmp (V)	2	2.02
Pmax (mW)	8.07	8.06
Rshunt (k-ohms)	0.05	0.055
R series (ohms)	36.5	36.74
Fidelity factor	0.739	0.69
Efficiency	0.42%	

Solar cell in dark

It is challenging to obtain reliable measurements when there are tiny variations in the light intensity since this creates a lot of noise into the system. Electrical approaches are used to introduce carriers into the circuit during dark IV measurements as opposed to light-generated carriers.

The first cell experimented under dark condition was poly-crystalline cell, the characteristics are as follow,

1. Poly-crystalline cell

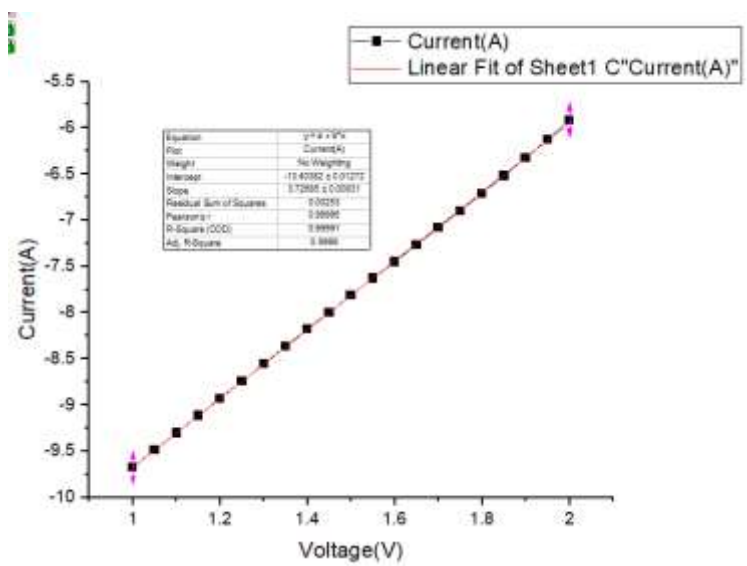


Figure 23 Dark I-V characteristics of poly-crystalline solar cell

We can observe from the graph, the IV characteristics of the poly-crystalline cell, this is a linear relation between the current and the voltage on the linear scale. We can obtain the reverse saturation current (leakage current) and the ideality factor from the IV curve.

Theoretically, the slope of the graph determines the ideality factor of the solar cell, to obtain the slope and the intercept of the curve we need to perform linear fitting of the curve for this cell, from the graph the slope is 3.728, so, applying the formula for slope we get,

$$m = \frac{40}{n} = \frac{40}{3.7268} = 10.73$$

As we know, ideality factor determines how closely the cell follows the diode equation and it should ideally be 1, and for a good diode it should be as close to 1, but for this cell it is determined to be 10.73, which is not very pleasant value for a poly-crystalline cell.

Another parameter to be determined from the IV graph is the reverse saturation current or the leakage current, this can be calculated with the help of the intercept value from the graph,

Leakage current, $I_0 = Antilog(Intercept)$

The intercept is 13.40

$I_0 = Antilog(-13.40) = 1.51\mu A.$

The leakage current for a cell should be as low as possible, but for this cell we can see that the leakage current is found to be quite high than expected from a poly-crystalline solar cell.

2. Mono-crystalline solar cell

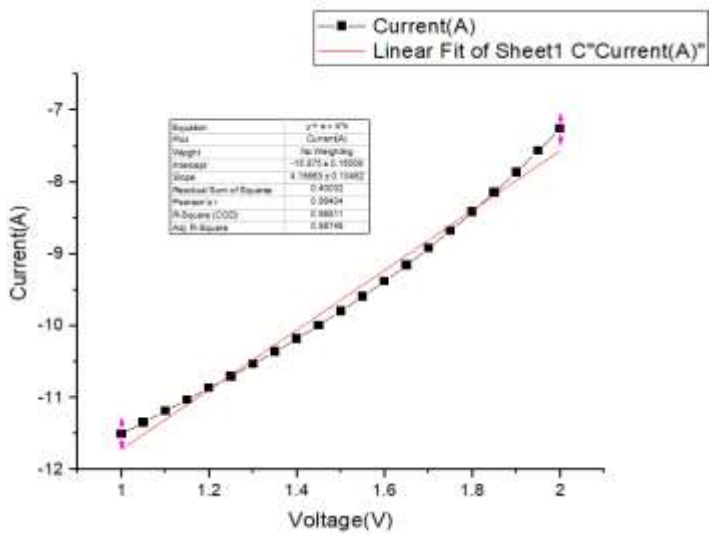


Figure 24 Dark I-V characteristics of Mono-crystalline solar cell

For the mono-crystalline cell, we can observe from the graph, it is a linear relationship between the voltage and the current, the slope for this graph is found to be 4.15,

Hence, the ideality factor for this cell will be, $m = \frac{40}{n} = \frac{40}{4.15} = 9.63.$

The ideality factor for mono-crystalline cell is better than poly-crystalline cell, it shows that the mono-crystalline cell follows the diode equation more closely than the poly-crystalline cell. For the leakage current the intercept from the graph is found out to be -15.87.

$I_0 = Antilog(-15.87) = 128.36\ nA.$

The leakage current for the mono-crystalline cell is very low, which is a desired characteristic for a solar cell.

3. Amorphous silicon solar cell

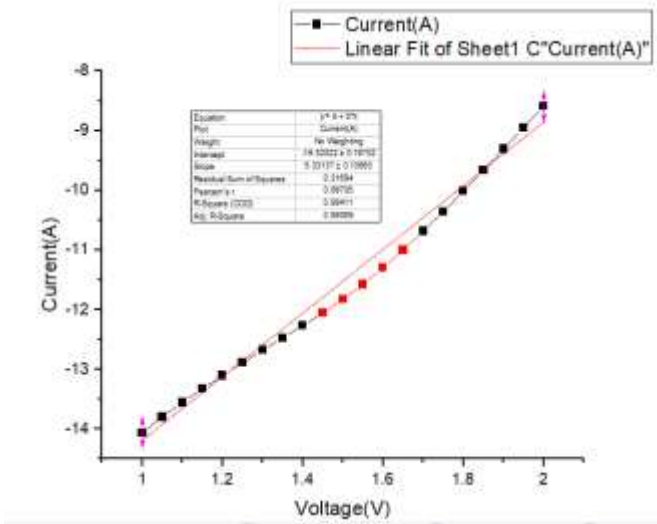


Figure 25 Dark I-V characteristics of amorphous silicon solar cell

We can observe a similar pattern for the amorphous silicon cell, after linear fitting of the graph, the leakage current and the ideality factor are found to be,

The slope from the graph was obtained as 5.29, so, the ideality factor will be,

$$m = \frac{40}{n} = \frac{40}{5.29} = 7.56$$

The ideality factor of the amorphous cell is good.

For the leakage current for this cell, the intercept found is -19.53, so, the leakage current is,

$$I_0 = Antilog(-19.53) = 3.3\ nA.$$

We can observe from this data that the leakage current for the amorphous cell is very low which is desirable.

C-V measurements

1. Poly-crystalline cell

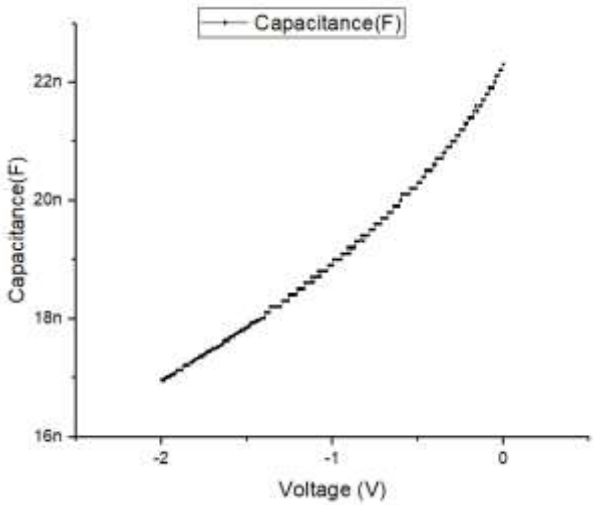


Figure 26 CV curve of a poly-crystalline cell

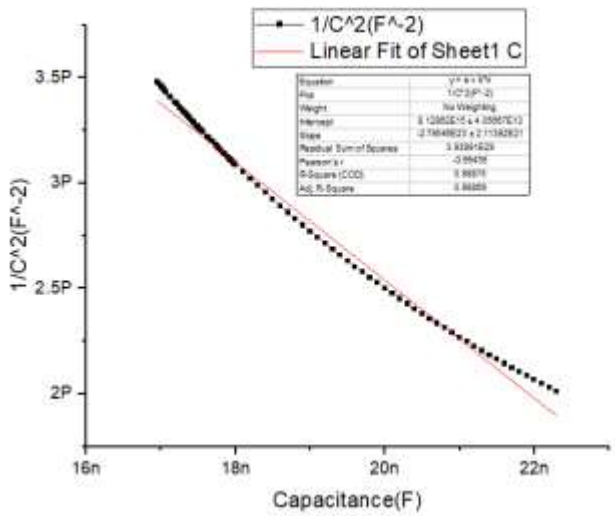


Figure 27 1/C^2 vs V curve for poly-crystalline cell

As we can observe from the graph, the CV curve is a linear relation, which is a desirable characteristic for the poly-crystalline cell, this shows that the depletion layer capacitance is good for this cell, the CV graph is plotted to check the junction capacitance of the cell, which is favorable for this cell.

The 1/C^2 vs V, graph is plotted to check the barrier height of the cell, linear fitting for this graph is done, and the barrier height for this cell was found to be 2.79/4, which equals to 0.697 eV, which is a typical junction capacitance value for a solar cell, so we can say that the cell is showing good charactersistics.

2. Mono-crystalline cell

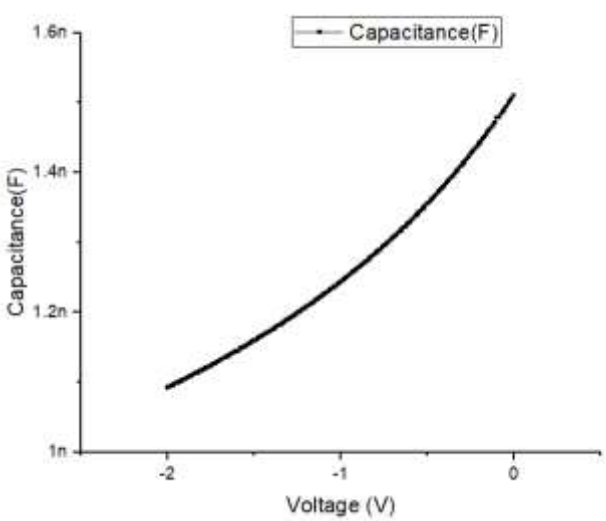


Figure 28 CV curve of a mono-crystalline cell

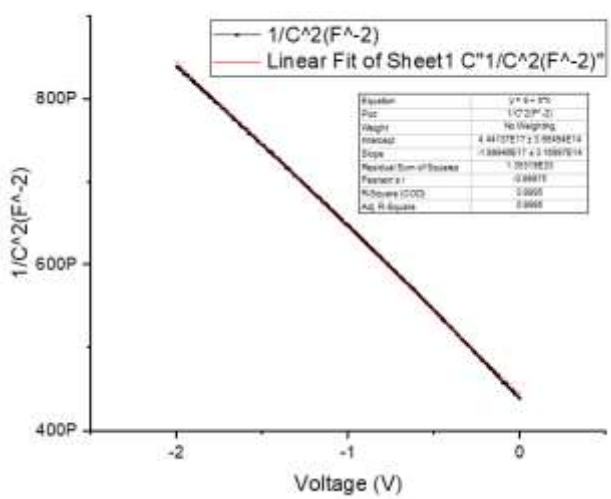


Figure 29 1/C^2 vs V curve for mono-crystalline cell

For the mono-crystalline cell, in the CV curve we found this curve not as linear as the poly-crystalline cell, but it shows a good behavior overall, this indicates that the depletion layer capacitance or also known as reverse bias capacitance is good for this cell.

From the linear fitting of the curve in figure 29, the barrier height calculated was 2/4 = 0.5 eV, which is also a desirable value for the barrier height of a solar cell, hence, we can say from this calculation that mono-crystalline cell shows a good behavior in dark conditions as well.

3. Amorphous silicon solar cell

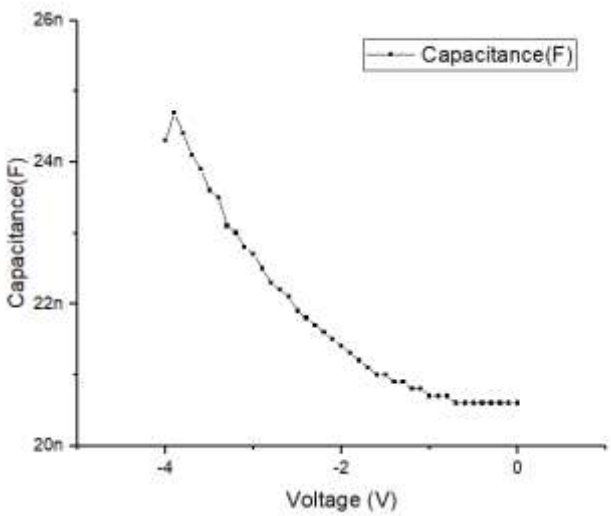


Figure 30 CV curve of amorphous cell at 1000 kHz

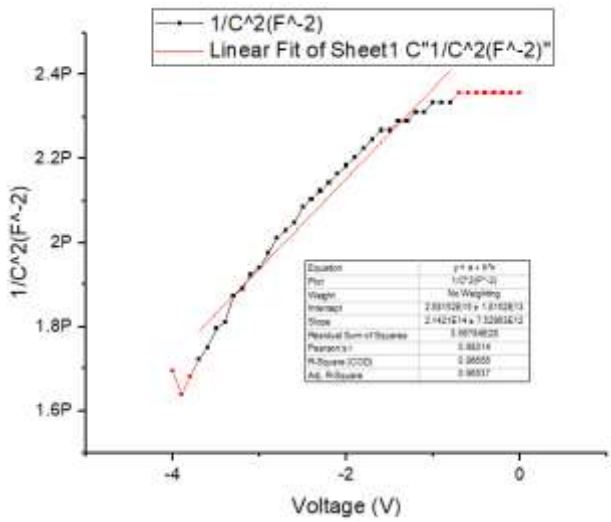


Figure 31 $1/C^2$ vs V curve amorphous cell at 1000 kHz

For the amorphous silicon cell, when the test was performed at the same frequency as the other two cells at 1000 kHz, but the behavior of amorphous cell at this frequency was found to be abnormal, even though the IV curve for this cell were showing favorable characteristics, the CV curve were not good.

Also, the $1/C^2$ vs V curve for amorphous cell, showed a completely flawed behavior, with a barrier height which cannot be calculated, so, from this we can say that the cell won't function normally at this high frequency.

So, another test was performed by lowering the frequency at 100 kHz, but the cell showed a similar behavior, which was not favorable, so, a further reduction of frequency was done, now, we conducted the test at 10 kHz, the characteristics curve are shown in figure below,

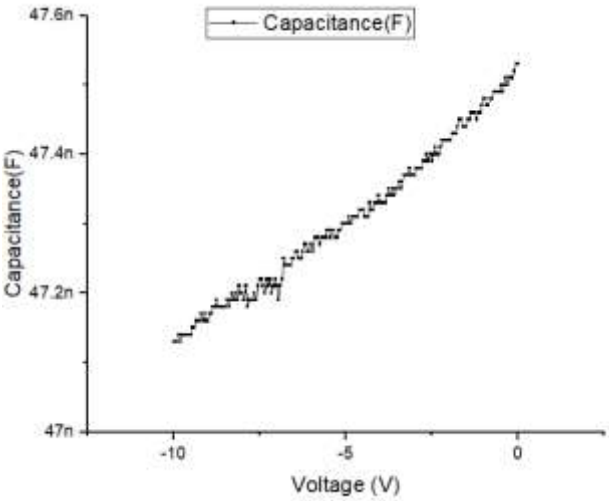


Figure 32 CV curve of amorphous cell at 10 kHz

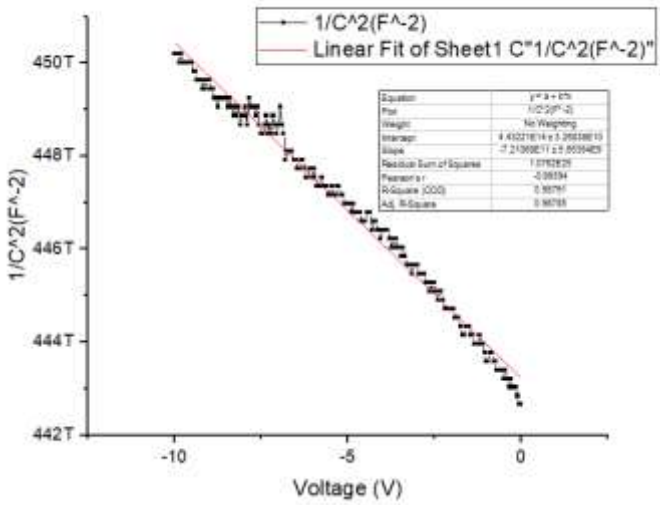


Figure 33 $1/C^2$ vs V curve for amorphous cell at 10 kHz

From this curve at 10 kHz, we finally can see an approachable behavior for this cell, it was still not the desired outcome for the cell, we found a linear behavior for CV curve, and for the barrier height, it was still very high than expected it was around 611/3 = 203.3 eV, which is very high and which could mean that the junction capacitance of this diode is broken.

This cell showed a working behavior at 10 kHz, from which we can deduce that amorphous silicon cells can be used in indoors than the outdoor due to it's non workable behavior at high frequency.

Impedance Spectroscopy

The aim of this experiment is to categorize the cells on the basis of the impedance spectroscopy, various tests are performed on the cell and different parameters were calculated for each cell in different conditions, now, on the basis of the impedance behavior the cell will be categorized.

1. Poly-crystalline solar cell

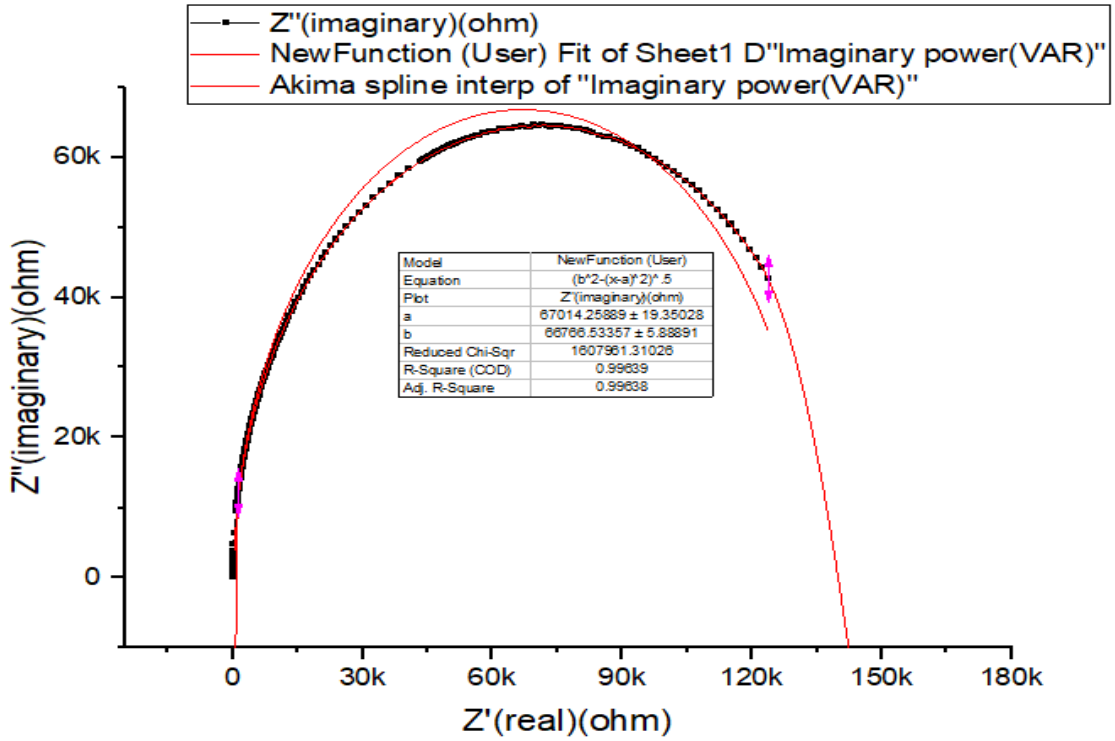


Figure 34 Impedance curve of poly-crystalline cell (in black) and extrapolated till X-axis for a curve fit (in red)

From the above curve for impedance spectroscopy for a poly-crystalline cell, we can observe that it follows the path of an ideal solar cell, which makes a complete semicircle, for the poly-crystalline cell we can see that the semicircle is not completed, the impedance spectroscopy is used to characterize the forward bias junction of the cell, so, by observing the above figure, we can says it has a good forward bias junction. For this curve after extrapolating the points to intersect the x-axis the point at which it intersects is 147.5 kohms. Although, we need to calculate the capacitance from this curve as well as the Zmin and w for the same.

We performed a curve fitting for this semicircle, by using the equation of the semicircle,

$$Y_0 = (b^2 - (x-a)^2)^{1/2}$$

Here 'a' is the point on Z"(imaginary) axis, where the centre of the semicircle lies, b is the point on the real axis from the origin of the plot. For the poly-crystalline cell the values of a and b were found to be,

a = 67014.25889 , b = 66766.5335, so for the value of capacitance we can write,

$$b = \frac{1}{2 * pi * f * C} = 66766.5355$$

The frequency here is the point in the curve where the centre of the semicircle is, which is 62 Hz for this graph,

Therefore , $C = \frac{1}{2*pi*62*66766.5355} = 3.84 * 10^{-8} F.$

Now, as we have the value of capacitance we can calculate the value of W at Zmin, which is derived from the ideal circuit equation, we get the Zmin = Rshunt/2 = 0.65 kohms.

So, W at Zmin = $\frac{1}{C*Rshunt} = 20.03\ k$

Hence, we got the value of capacitance, W as well as Zmin, so we can say for the poly-crystalline curve, it shows a goo behavior, with typical values. The curve fit and the impedance curve for this cell is also following a good behavior, so, we can say that the poly-crystalline cell is a good solar cell.

2. Mono-crystalline cell

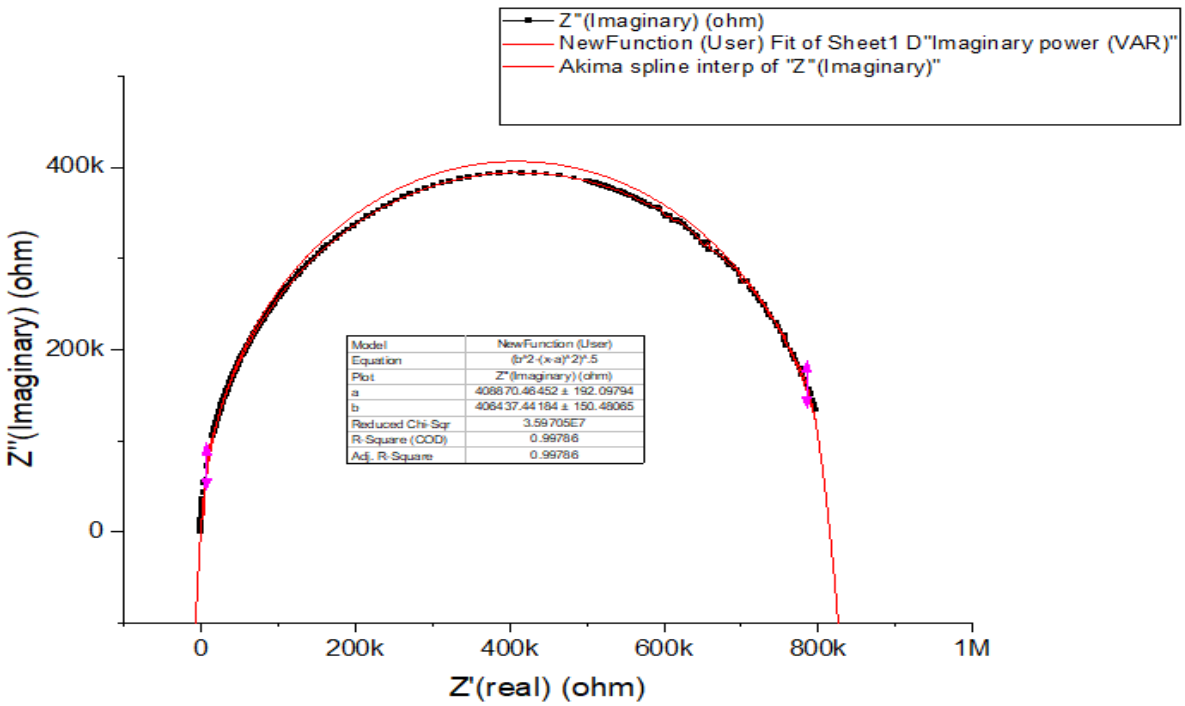


Figure 35 Impedance curve of Mono-crystalline cell (in black) and extrapolated till X-axis for a curve fit (in red)

From the above impedance curve for the mono-crystalline cell, we can tell that the graph shows good characteristics in the forward bias conditions, although, as we have measure the impedance values from 20 Hz till 1MHz, we can see that the plot is not touching the x-axis, for that we have extrapolated the further point until it intersects the X-axis, this graph intersects the x-axis at 832 k ohms.

The values for 'a' and 'b' for this curve are a = 408870.46 and b = 406437.4418, hence, we can calculate the capacitance value as,

C = $\frac{1}{2*pi*129*406437.44} = 3.03 * 10^{-9}F.$

We observe that the value of capacitance is nearly 3 nF, which is a good value for a capacitance, this capacitance value is better than the poly-crystalline cell, and we can also observe the impedance curve which is a better curve than that of the poly-crystalline cell. The value of Zmin = Rshunt/2 = 2.36 kohms.

Now as we have the values of capacitance and shunt resistance we can calculate the value of W at Zmin, which can be given as,

$$W\ at\ Z_{min} = \frac{1}{C * Rshunt} = 70k.$$

Hence, we obtained all the parameters for the mono-crystalline cell from the impedance curve, and we can say that the mono-crystalline cell shows a very good behavior in forward bias conditions, and as the curve fit for this graph is also accurate, we can say that this mono-crystalline cell is a good cell.

3. Amorphous silicon cell

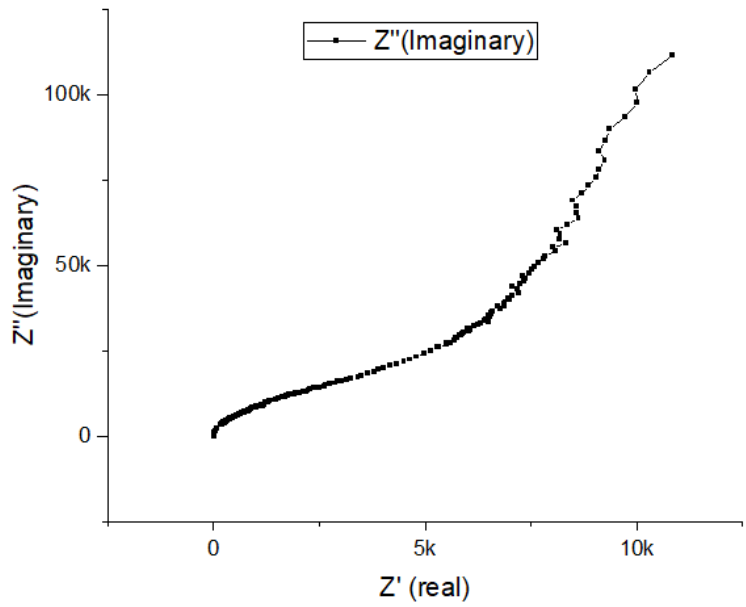


Figure 36 Impedance curve of amorphous silicon solar cell

The impedance curve for the amorphous cell is very poor, as there are no signs of semicircle at the first glimpse, and the relation between the imaginary and real impedance also looks very absurd, which is not a expected behavior from any type of solar cell, if we take a close look at this graph we can find two instances where we can say that it forms a semicircle, a small one but it does have the property of the semicircle, below are the graphs from which we can take our observations further,

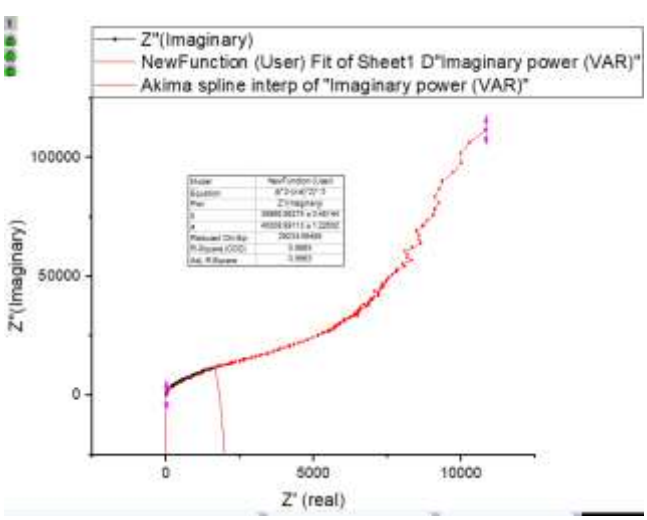


Figure 37 Curve fitting for the amorphous cell part 01

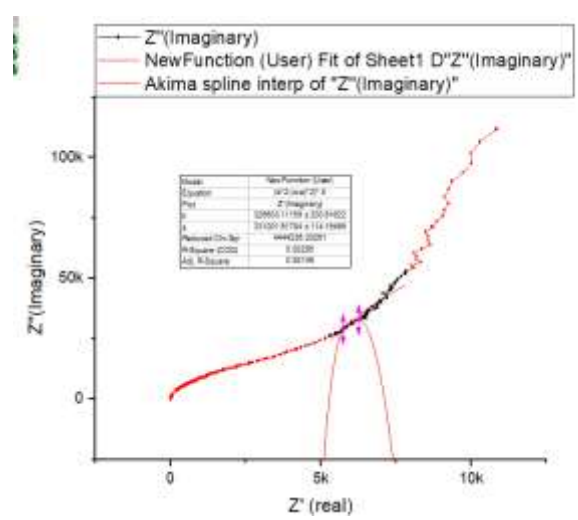


Figure 38 Curve fitting for the amorphous cell part 02

Further, simplification of the graph, we can plot two semicircles from this plot, the first one, at from the point of zero impedance and second one in between of the curve, we can now calculate all the parameters from this simplified plot which were performed for the previous two cells, starting with the part 01, we found the values of 'a' and 'b' as, a= 39966.86 and b = 40008.99, from this we can calculate the value of capacitance at the frequency 444 Hz,

C = 8.96 nF, we also calculated the value of Zmin as 0.025 kohms,

The value of W at Zmin = 2200k,

Now, as we observe these values, these parametric values are still not close to the typical values of a solar cell to be in a good condition, we can say that the junction of this cell is dodgy. Further we calculated the value of same parameters for the second part of the curve and they were,

a = 326603.111 , b = 331001.917, so the capacitance at the frequency 78 Hz will be,

C = 6.16 nF, and the Zmin will be the same for the entire cell, so we don't need different calculations for this parameter, although the W at Zmin will differ, it is now, Z min = 3240 k.

Still, when we split the plot into two different curves, we can see a little tendency of the cell to behave as a good cell, but it is far from acting like a proper good cell under normal conditions, so, we can say that the amorphous cell is not a good cell, by the calculations and the impedance spectroscopy of the cell.

Discussion:

After all the tests are conducted on each of the cells we can now decide and categorize these cells individually by inspecting their parameters and behavior, starting with the characteristics of cell under illumination, various parameters were calculated for the test of the cell under illumination, from which the important parameters which would determine the characteristics of the cell were, Fill factor, series and shunt resistance and the efficiency of the cell, these IV curves under light determines how optically configure the solar cell is under those conditions, and we found that poly-crystalline cell showed the best behavior out of the three cells which were tested, although the fill factor, which practically is the power conversion efficiency of any solar cell, was not as good as the other two cells, the efficiency of the poly-crystalline solar cell is more than the mono-crystalline cell and the amorphous cell. Whereas the mono-crystalline cell has the efficiency of 2.6%, which is lower than the poly-crystalline cell but is greater than the amorphous cell, we can say that the mono-crystalline cell, also have a good behavior, compared to amorphous cell, which has only an efficiency of 0.46%, which depicts a very poor behavior under illumination.

When we look at the series and shunt resistance for all of these cells, we can observe a very little difference between them for poly-crystalline and mono-crystalline when compared to the amorphous cell, which shows a very poor value. Hence, we can say that the amorphous cell has a very poor conversion rate as well as overall poor behavior.

Next, when we take a look at the dark IV and CV curves for all three cells, we check the leakage current and the ideality factor of the cell, these dark IV curves gives the extra information about the cell for more diagnostics purpose, when calculated the reverse saturation current (leakage current) we observed that the poly-crystalline cell, which had the best efficiency out of the three cells, has the most leakage current compared to the other two cells, which is a poor conduct, as this shows that the poly-crystalline cell has some initial high voltage stress which leads to high leakage current, while for the mono-crystalline and amorphous cell, the leakage current was found to be as low as in nano-amperes. Another factor which was calculated from the dark IV curve is the Ideality factor, for an ideal cell this value is 1, but for practical solar cell this should be as close to 1, this can be determined from the slope of the graph, and it is found that poly-

crystalline cell has the worst value of ideality factor from the three cells, it is 10.73, while the ideality factor is 9.63 and 7.56 for the mono-crystalline and amorphous cell respectively. It is to the notice that amorphous cell despite having a very poor illumination characteristics it shows the best dark IV characteristics than the other two cells.

Moving onto the CV curve for the cells, we can check how good the cell is under reverse bias and calculate the barrier height, and by observing all three cells, we can say that the mono-crystalline cell has the lowest barrier height, compared to the other two cells, followed by the poly-crystalline cell and amorphous cell, amorphous cell has a very poor CV curve, it does not follow the typical behavior of the cell, and the barrier height of the amorphous cell is very high, despite amorphous cell showing better IV characteristics, the CV curve says otherwise, meaning the cell does not function properly under reverse bias, but when the frequency was reduced to 10 kHz, amorphous cell showed a working behavior, and a decent CV curve, from which we can say that this amorphous cell will be better working in indoor conditions rather than outdoor conditions. It has the worst CV characteristics. So, overall we can say that mono-crystalline cell has the best behavior under dark conditions followed by ploy-crystalline cell and amorphous with the worst characteristics, but when we reduce the frequency, amorphous cell shows a better behavior.

When we observe the impedance curves for all cells, we calculated the capacitance, W at Zmin and Zmin for all the cells, and from the calculated values and the curve behavior of the cell, we can say that mono-crystalline cell has the better fit and parametric value than the other two cells, starting with the capacitance of the cells, mono-crystalline cell has capacitance of 3.3 nF/cm^2, which is lower than a typical value of the cell, while we observe the curve fitting of the cell, we can find that the mono-crystalline cell has a better semicircle than the other two cells, coming to poly-crystalline cell, the curve for this cell was good, but not with the best behavior, although the capacitance for the poly-crystalline cell was near to the typical value which was found to be 37.8 nF/cm^2, we can say that poly-crystalline cell shows a good behavior in terms of curve fitting.

For amorphous silicon solar cell, we observed from the graph, that this cell does not show any working behavior under normal environmental conditions, but from the graph, we actually deduced two semicircles, from which we managed to do a curve fitting for this cell, and we found the capacitance value for both these curves, and to the surprise, these values were not as poor as expected, they were 10.2 nF/cm^2 and 365 nF/cm^2, which are reasonable values for a cell showing a very poor behavior, so from these observations, we can say that the amorphous cell is a poor cell but if it is kept under certain conditions, this cell can work as a normal solar cell.

Conclusions

References:

[1]. Pas.rochester.edu. 2022. *The Solar Spectrum*.

[2]. Yan, J. and Saunders, B., 2014. Third-generation solar cells: a review and comparison of polymer:fullerene, hybrid polymer and perovskite solar cells. *RSC Adv.*, 4(82), pp.43286-43314.

[3]. Binz, C., Tang, T. and Huenteler, J., 2017. Spatial lifecycles of cleantech industries – The global development history of solar photovoltaics. *Energy Policy*, 101, pp.386-402.

[4]. Aps.org. 2022. *This Month in Physics History*.

[5]. Encyclopedia Britannica. 2022. *solar cell | Definition, Working Principle, & Development*. [6]. *Solar Energy Materials*, 1990. Thermal storage materials solar energy materials special issue. 21(1), p.91.

[7]. Techxplore.com. 2022. *Solar cells: Layer of three crystals produces a thousand times more power*.

[8]. Rau, U. and Kirchartz, T., 2019. Charge Carrier Collection and Contact Selectivity in Solar Cells. *Advanced Materials Interfaces*, 6(20), p.1900252.

[9]. Guerra, N., Guevara, M., Palacios, C. and Crupi, F., 2018. Operation and physics of photovoltaic solar cells: an overview. *I+D Tecnológico*, 14(2), pp.84-95.

[10]. Kranthiraja, K., Gunasekar, K., Kim, H., Cho, A., Park, N., Kim, S., Kim, B., Nishikubo, R., Saeki, A., Song, M. and Jin, S., 2017. High-Performance Long-Term-Stable Dopant-Free Perovskite Solar Cells and Additive-Free Organic Solar Cells by Employing Newly Designed Multirole π-Conjugated Polymers. *Advanced Materials*, 29(23), p.1700183.

[11]. Abdel-Ghany, A. and Al-Helal, I., 2014. Methods for determining the temperature of a plastic net under solar and thermal radiation conditions. *Solar Energy Materials and Solar Cells*, 125, pp.1-7.

[12]. Riordan, C., Hulstrom, R., Cannon, T. and Myers, D., 1991. Solar radiation research for photovoltaic applications. *Solar Cells*, 30(1-4), pp.489-500.

[13]. Bouzidi, K., Chegaar, M. and Aillerie, M., 2012. Solar Cells Parameters Evaluation from Dark I-V Characteristics. *Energy Procedia*, 18, pp.1601-1610.

[14]. Rappaport, P., 1959. The photovoltaic effect and its utilization. *Solar Energy*, 3(4), pp.8-18.

[15]. Hoenig, R., Glatthaar, M., Clement, F., Greulich, J., Wilde, J. and Biro, D., 2011. New measurement method for the investigation of space charge region recombination losses induced by the metallization of silicon solar cells. *Energy Procedia*, 8, pp.694-699.

[16]. "Solar cells—Operating principles, technology and system applications", *Solar Energy*, vol. 28, no. 5, p. 447, 1982.

[17]. "Dark IV Measurements | PVEducation", *Pveducation.org*, 2022.

[18]. M. Müller, "Reporting Effective Lifetimes at Solar Cell Relevant Injection Densities", *Energy Procedia*, vol. 92, pp. 138-144, 2016.

[19]. M. Suresh, "Measurement of solar cell parameters using impedance spectroscopy", *Solar Energy Materials and Solar Cells*, vol. 43, no. 1, pp. 21-28, 1996.

[20]. T. Aburaya, "Analysis of 10 years' flight data of solar cell monitor on ETS-V", *Solar Energy Materials and Solar Cells*, vol. 68, no. 1, pp. 15-22, 2001.

[21]. E. von Hauff and D. Klotz, "Impedance spectroscopy for perovskite solar cells: characterisation, analysis, and diagnosis", *Journal of Materials Chemistry C*, vol. 10, no. 2, pp. 742-761, 2022.

[22]. T. Ibn-Mohammed et al., "Perovskite solar cells: An integrated hybrid lifecycle assessment and review in comparison with other photovoltaic technologies", *Renewable and Sustainable Energy Reviews*, vol. 80, pp. 1321-1344, 2017.